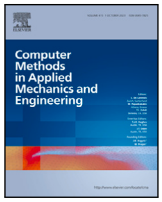
Computer Methods in Applied Mechanics and Engineering 442 (2025) 118044

Contents lists available at ScienceDirect

Comput. Methods Appl. Mech. Engrg.

journal homepage: www.elsevier.com/locate/cma



Accelerating cell topology optimisation by leveraging similarity in the parametric input space

A. Martínez-Martínez [a,*], D. Muñoz [b], J.M. Navarro-Jiménez [a], O. Allix [c], F. Chinesta [b], J.J. Ródenas [a], E. Nadal [a]

[a] *Institute of Mechanical and Biomechanical Engineering, Universitat Politècnica de València, Spain*
[b] *PIMM Laboratory, Arts et Métiers Institute of Technology, CNRS, France*
[c] *ENS Paris-Saclay, CNRS, LMT – Laboratoire de Mécanique et Technologie, Université Paris-Saclay, France*

ARTICLE INFO

Keywords:
Machine Learning
High-resolution Topology Optimisation
Finite Element Analysis
Trabecular structures
Metric definition
Parametric space exploration
Data acquisition

ABSTRACT

The design of high-resolution topology-optimised (TO) structures is important for many industrial and medical applications because of their better mechanical performance under different load conditions. Traditional density-based TO methods, like the Solid Isotropic Material with Penalisation (SIMP) method, can produce detailed designs but are very computationally expensive, especially for fine meshes. While surrogate models using neural networks can speed up the process, they often lack generality and can create discontinuities, making them less effective for solving new problems.

This study addresses these issues by introducing a method to speed up cell-level TO within a 2-Level framework, where large structures are built by combining optimised square cells. A data-driven instance-based model provides a better starting point for the standard SIMP-based optimiser, placing it closer to a local minimum and reducing computation time. To avoid the generality problems of other methods, the instance-based model uses a dataset expanded through two strategies: context-based data creation, which generates specific samples for the problem, and data augmentation, which increases dataset size without extra computation.

Two similarity metrics, vector-based and energy-based, are used to measure how close the input parameters are. Both metrics are effective, but the energy-based metric is expected to work better in 3D cases, where higher-dimensional input spaces make other approaches less reliable. This methodology addresses important challenges associated with existing instance-based models, enhancing the speed and applicability of high-resolution TO.

1. Introduction

Structural topology optimisation (TO) has become crucial in industrial design [1] for its ability to optimise material distribution while fulfilling specified constraints within a design domain. In structural problems, a common application involves minimising structural compliance under a volume fraction constraint.

In implant design, TO provides interesting structural characteristics to improve conventional designs. For instance, topology-optimised implants help to prevent stress shielding [2], which is the change in bone density due to the change in the load transfer inside the bone when the implant is placed. Additionally, TO techniques can provide high-resolution or multi-level structures with trabecular distributions of bars that perform well under unexpected load conditions [3]. These complex structures can be

* Corresponding author.
E-mail address: anmar29a@etsid.upv.es (A. Martínez-Martínez).

https://doi.org/10.1016/j.cma.2025.118044
Received 4 January 2025; Received in revised form 21 April 2025; Accepted 21 April 2025
Available online 9 May 2025

obtained using additive manufacturing and consist of an infill formed by multiple small bars [4–6], similar to the porous nature of bone structures [1]. This could be particularly interesting for bone tissue engineering, which designs treatments for critical-sized bone defects [7]. In this medical context, defining an interconnected porous structure with a specific porosity and porous size is necessary to enhance bone ingrowth [8]. There are two main strategies for generating these trabecular structures: lattice optimisation introduces holes with a clear repetitive pattern [9,10], whereas density-based methods introduce holes without a distinct pattern. The former group provides computationally efficient solutions but lacks topology variability, potentially reducing the structural properties of the final design. In contrast, density-based methods are characterised by significant topological versatility.

One of the most popular density-based TO approach is the Solid Isotropic Material with Penalisation (SIMP) method based on finite element method (FEM) calculations [11,12]. For convenience, we refer to it as FEM-SIMP. However, a notable limitation of the traditional SIMP method lies in the strong correlation between the size of geometrical features and the discretisation resolution. Consequently, achieving high-resolution structures necessitates using fine meshes, leading to important computational costs [13].

To overcome this challenge, multi-level methods [14] have been proposed. Nonetheless, their primary limitation lies in the lack of structural continuity. In response, the 2-Level optimisation strategy proposed in [15] aims to reduce computational costs while maintaining structural continuity significantly. In this approach, TO is initially solved at coarse discretisation. Once the coarse-level TO problem is solved, the discontinuous FE stress field is post-processed to obtain a distribution of equilibrated tractions acting on the boundaries of the elements of cells. These equilibrated tractions satisfy the equilibrium on each cell and represent the continuity of tractions (equilibrium) between adjacent elements. Then, each cell is discretised again, and a new TO problem emerges in each cell of the macro problem domain. This finer-scale TO problem is characterised by a square-domain definition of all the cells, requiring the solution of individual cell TO problems. Despite reducing the demand for computational resources, time-sensitive applications like medical simulations demand even more efficient computations.

The use of data-driven surrogate models (DDTO) has emerged as a possible acceleration strategy in the context of TO. Most of these approaches rely on training a neural network (NN) using a dataset previously computed using FE calculations, which presents multiple limitations [16–22]. The primary limitation of these approaches lies in their lack of generality [23,24]. Particularly, surrogate models' performance is poor when solving new one-level problems featuring different domains and boundary conditions [25]. Additionally, due to being data-driven, structural discontinuities may appear in the final outcome. As a result, DDTO remains an open field for further investigation, especially in the context of real-world applications under realistic circumstances [26]. Our proposal is focused on the 2-Level TO presented in [15], mitigating most of the previously mentioned limitations. Here, a macro structure is discretised in cells with square domains of uniform sizes. This characteristic leads to independent cell's TO problems without changes in the geometric definition. This is particularly helpful for the introduction of ML techniques.

Therefore, the main goal of the proposed method is to accelerate the optimisation at the cell level. Here, an instance-based model proposes an initial guess for the FEM-SIMP optimiser closer to the local minima than the standard approach, an initial uniform density distribution. Unlike surrogate models such as NN, this instance-based model directly proposes previously optimised cells from a dataset to enhance the optimisation of subsequent problems.

The implementation of an instance-based model at the cell level allows for overcoming most of the previous limitations. Firstly, like the original method, it can be applied to any macrostructure represented as an assembly of cells. Secondly, parameter space exploration and dataset creation are more efficient. A meaningful and extensive database is generated by leveraging existing problems or designing ad-hoc ones and applying the two-level optimisation procedure. Optimised cells, of which there are many, are added to this database and can later serve as starting candidates for solving new optimisation problems. Furthermore, optimised cells can be incorporated into the dataset on the fly, even before completing the global topology optimisation. Thus, each new computation initially performed online becomes part of the offline dataset, improving efficiency for future optimisation tasks.

This approach aligns with previous studies that have employed ML models to propose improved initial guesses, thereby accelerating iterative optimisation processes [27,28]. Particularly, instance-based models are used in the context of incompressible flow problems to enhance initial guesses for iterative solvers, leading to faster convergence [27].

One of the main challenges of this approach is the large number of parameters defining the cell's TO problem, including the traction field on the edges and the volume fraction. This complexity can lead to difficulties in measuring distances between cases and requires a substantial amount of data to train surrogate models.

In 2D, it appears that using an Euclidian distance in the parameter space of dimension 17 still allows us to select, in most cases, a proper initial guess. Nevertheless, we explore the possibility of using a reduced and mechanically meaningful metric based on the strain energy of the macro cell whose interest is to measure distance in a 4D space for the 2D problem and in a 7D space in 3D. This allows avoiding the lack of sensitivity of the distance we expect to arise in 3D optimisation problems.

The additional problem is that covering a space of high dimension with data is highly challenging and demanding. To circumvent the problem, we reduce the exploration to the region of interest associated with the existing optimisation problem. Then, it might be possible to fill this region of interest by performing new two-level optimisation problems or/and to proceed to a simple data augmentation strategy consisting of rotating the cell, which multiplies the number of available data by 8 in 2D and 24 in 3D with no computational cost.

This paper is structured as follows. Section 2 presents the baseline methods from the literature used in this study. First, the TO problem and the 2-Level methodology are explained in Sections 2.1 and 2.2, respectively.

Section 3 describe the contributions of this work. In Section 3.1, the cell TO problem is examined from a machine-learning perspective. Then, Section 3.2 details the dataset creation process, including the proposed macro problems and two data generation strategies. Sections 3.3 and 3.4 then introduce the methodology for accelerating cell TO and the metrics used to measure cell similarity.

Next, Section 4 presents the numerical results, analysing the speed-up effects of different data creation strategies and distance definitions. Section 5 discusses the acceleration results and the particular characteristics of the proposed methodology. Finally, Section 6 summarises the main conclusions of this work.

2. Baseline methods

This section presents previously developed methodologies by other authors and in our group that serve as the foundation for the contributions of this paper. First, the TO problem (see Section 2.1) and the 2-Level TO method [15] (see Section 2.2) are briefly introduced. The key characteristics of these methods are highlighted to justify the introduction of acceleration strategies in the cell-level TO process while preserving the generality of the approach.

2.1. Topology optimisation problem - SIMP method

TO methods in structural problems aim to find the optimal material distribution in a given design domain Ω, adjusting the relative density ρ of each mesh element. In this application, the objective function to minimise is the compliance c, which is equivalent to maximising the component's stiffness. The SIMP method [29–31] defines a volume fraction restriction to limit the amount of material used in Ω. This approach considers ρ as a continuous variable with values in the range $[0, 1]$. Additionally, a penalisation parameter p enforces the segregation between material ($\rho = 1$) and void ($\rho = 0$) and penalises intermediate density values. As a result, the SIMP method formulates the TO as follows

$$
(\mathbb{TO}(v_f)) = \begin{cases}
\min\limits_{\rho,\mathbf{u}} : & c(\rho;\mathbf{u}) = \dfrac{1}{2}\displaystyle\int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{u})\mathbf{D}(\rho)\boldsymbol{\varepsilon}(\mathbf{u})d\Omega & \text{(1a)}\\
& \text{with } \mathbf{D}(\rho) = \rho^p \mathbf{D}_0 & \text{(1b)}\\
\text{subject to:} & V(\rho) = \displaystyle\int_{\Omega} \rho d\Omega = \overline{V}_e \sum \rho_e = v_f V_0 & \text{(1c)}\\
& a(\mathbf{u},\boldsymbol{v};\rho) = l(\boldsymbol{v}) \text{ where,} & \text{(1d)}\\
& a(\mathbf{u},\boldsymbol{v};\rho) = \displaystyle\int_{\Omega} \boldsymbol{\varepsilon}(\mathbf{u})\mathbf{D}(\rho)\boldsymbol{\varepsilon}(\boldsymbol{v})d\Omega & \text{(1e)}\\
& l(\boldsymbol{v}) = \displaystyle\int_{\Omega} \mathbf{b}^T \boldsymbol{v} d\Omega + \int_{\Gamma_N} \mathbf{t}^T \boldsymbol{v} d\Gamma_N & \text{(1f)}\\
& 0 \leq \rho_{min} \leq \rho \leq 1, & \text{(1g)}
\end{cases}
$$

where $\boldsymbol{\varepsilon}$ and $\mathbf{u}$ represent the strain and displacement fields, respectively, and $\boldsymbol{v}$ is the virtual displacement field. $\mathbf{D}$ is Hook's law matrix, which relates the strains and stresses. Moreover, v_f is the predefined volume fraction, V_0 is the total volume of Ω and V_e is the volume of the element e.

A low-pass filter is applied to the compliance sensitivities to avoid numerical instabilities in the density distribution, such as checkerboard patterns [32]. The optimality criteria (OC) algorithm updates the density distribution based on the filtered compliance sensitivities. We adopt a common filtering approach where the density value is averaged over neighbouring elements within 1.5 times the element size from the center of each element. However, other optimisation algorithms could also be used. More details of the implementation are explained in [32].

2.2. Individual cell-level optimisation in a two-level topology optimisation approach

In this work, the methodology to obtain locally-optimised trabecular structures is the 2-Level TO process presented in [15], where two levels of discretisation are used. There are two main stages in this multi-level process: the density optimisation at the coarse level and the individual optimisation of the cells (see Fig. 1).

In this method, a cell is a square subdomain into which the macro problem is divided. At this higher level, each cell may contain one or more elements of the macro-level mesh, allowing for future implementations with multiple elements per cell. However, in this work, each cell is always equivalent to a single element in the macro-level mesh.

A notable difference between these stages lies in using the penalisation factor, introducing distinct behaviours based on their magnitude. When the penalisation factor approaches one, it facilitates a continuous distribution defined by intermediate density values, resulting in a structure without sharp boundaries [33]. Conversely, an increase in the penalisation factor accentuates the differentiation between regions identified as material (solid) and void (empty space), providing a well-defined structure of bars.

In the context of one-level TO approaches, the continuous distribution (low penalisation) is considered undesirable as intermediate values are impractical for manufacturing processes. However, in the 2-Level TO employed in our study, both penalisation scenarios find application to generate the final trabecular structure: penalisation $p = 1$ will be used in the global problem optimisation to obtain a low-definition/coarse material distribution, then $p = 3$ will be used at the cell level to create the final high-resolution material distribution.

Additionally, the final trabecular structure is mainly determined by the discretisation size used in the cell-level stage, independently from the rest of the cells. Consequently, trabecular feature size can be controlled if necessary with the fine mesh size.

Regarding the substeps followed at each stage, firstly, the coarse-level topology optimisation procedure is similar to the conventional implementations as follows:

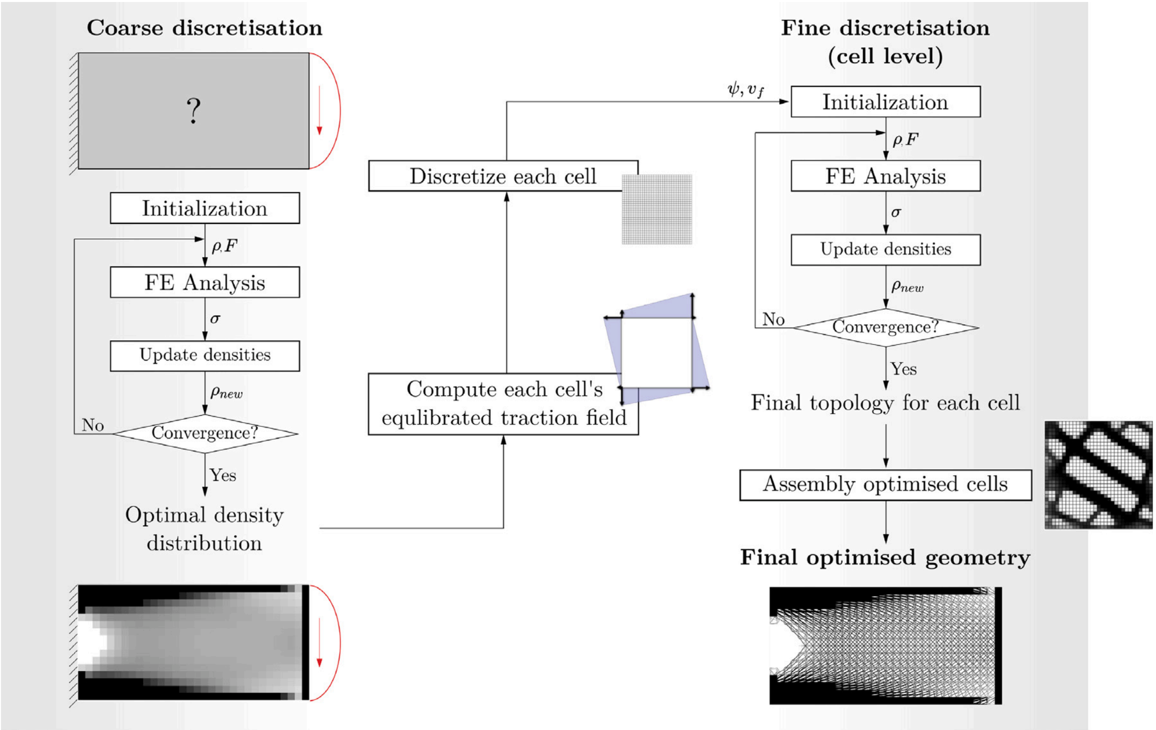


Fig. 1. Two-level TO methodology proposed in [15] consisting on the macro optimisation problem and the cell-level optimisation. Particularly, the TO of the fine discretisation (marked in the grey area) is the basis of the acceleration strategy presented in this work.

1. Initialise the problem variables, such as boundary conditions, volume fraction and material properties.
2. Run FE analysis to solve the elasticity problem and obtain c.
3. Update densities. This step can be done with different strategies. In this work, we use the SIMP scheme with a low penalty value ($p = 1$).
4. Check if convergence is reached. In that case, the result is the final coarse optimised topology. Conversely, the loop is repeated from point 2 using the new density field.

This step is computationally affordable because the number of degrees of freedom is reduced in the coarse discretisation. Once the coarse-level problem is finished, the cell-level topology optimisation starts. Given the low penalty value imposed in the coarse-level TO, the optimal density distribution is defined by void and full cells with extreme density values alongside multiple with intermediate-density values. While such solution may not be desirable in a single-level optimisation, within the framework of the 2-Level TO, it is addressed in the second stage of the multi-level discretisation strategy. Thus, only the intermediate-density cells from the coarse-level problem become individual TO problems, while cells with extreme values maintain their definitive status. The number of intermediate-density cells can be controlled using a density threshold, as demonstrated in [15]. After the coarse stage, cells with high and low-density values are directly assigned as full or void cells, respectively. Subsequently, the equilibrated tractions applied on each cell are computed from the solution of the coarse problem, effectively representing the boundary conditions of each new TO cell problem. Moreover, each cell domain is discretised using a uniform Cartesian grid, and the intermediate-density value from the coarse TO solution defines the volume fraction constraint of each cell TO. Once this intermediate preprocess is finished, the cell-level TO procedure follows these steps:

1. Initialise the problem variables. The equilibrated tractions applied on each cell have previously been computed. Additionally, the predefined volume fraction is determined by the relative density of the cell obtained at the end of the coarse optimisation process.
2. Run FE analysis to solve the elasticity problem and obtain c.
3. Update new densities using SIMP method with a high penalty value ($p = 3$) to obtain solid or void density values and a well-defined optimised cell's geometry.
4. Check if convergence is reached. In that case, the result is the final optimised topology. Conversely, the loop is repeated from point 2 using the new density field.

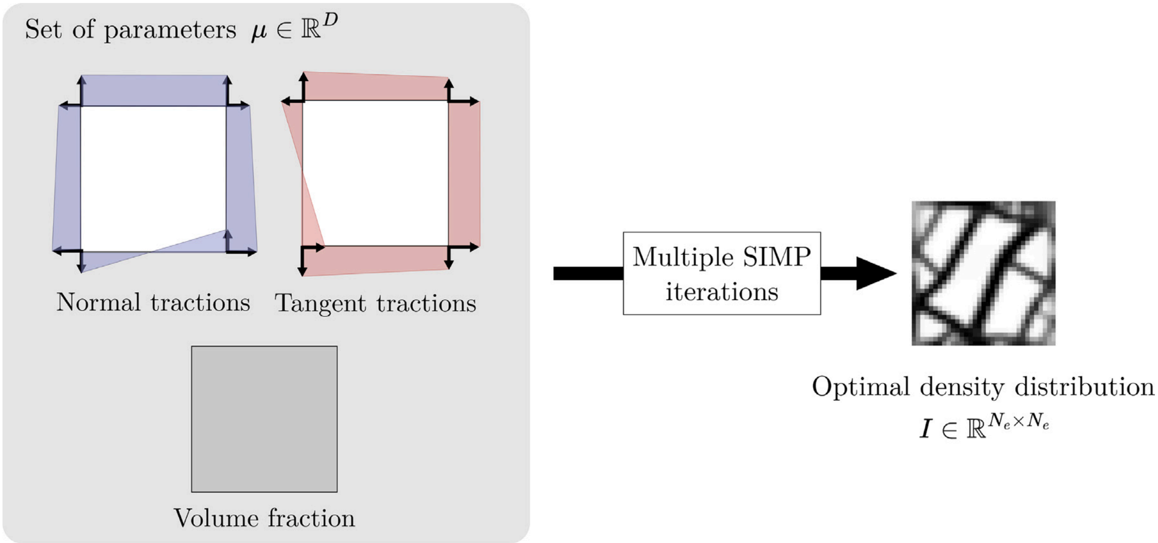


Fig. 2. The one-shot prediction strategy proposes a instance-based model model which projects from the parameters space (μ) to the geometry space (I). The parameter space is formed by boundary conditions and volume fraction. The geometry space is defined by an image of density distribution.

This cell-level TO process is parallelisable, and the final optimised geometry of the global problem is obtained by assembling multiple optimised cells. However, the most significant computational cost lies in the cell-level TO problem, primarily due to the multiple SIMP iterations and FE analyses required for convergence.

3. Methodology

This section describes the contributions and methodologies used in this paper. First, we present the problem under study: cell-level TO. Section 3.1 describes the cell-level TO process from a ML perspective, since speed-up strategies and dataset creation are only focused on this cell-level stage. In Section 3.2, we detail the structural problems used to generate our dataset from their optimised cells. Then we introduce two data creation strategies designed to increase the available information for the presented global problems.

Finally, Section 3.3 presents the acceleration procedure for the 2-Level TO, which involves as initial guess, the density distribution of a previously-optimised cell in the dataset. The new density distribution will be the closest to the query sample according to a chosen distance metric. Since different distance definitions can influence the selection process, Section 3.4 introduces two distance metrics for measuring cell similarity.

3.1. *Definition of cell's TO problem spaces*

As presented in Section 2.2, the coarse discretisation TO is related to the global structural problem (structure geometry and boundary conditions), contrary to the cell-level TO, which consists on individual TO problems. Once the coarse-level TO is finished, each cell TO problem is defined by a set of parameters μ which includes cell's traction distribution and volume fraction.

This work introduces ML techniques at the cell level to accelerate the process by leveraging several advantageous properties. By using the repetitive nature of the cell domain, which consists of squares, the design domain for cell-level TO is maintained consistently. Additionally, traction fields can be scaled between -1 and 1 without altering the final geometry of the optimised cell, as the solution depends only on the distribution of tractions, not their absolute values. Furthermore, these traction fields must satisfy the equilibrium equations, effectively reducing the dimensionality and size of the parameter space. Consequently, this approach narrows the search space at the cell level while retaining the capability to optimise any global problem by combining multiple optimised cells. Thus, the use of ML techniques in this framework is compatible with the versatility and capability of the original 2-Level TO method [15].

In this paper, only 2D structural problems are considered, and the traction field distribution is linear along the faces of the cell's domain, like in [15]. As a result, $\mu \in \mathbb{R}^D$ is defined by the 16 nodal coefficients of the boundary conditions (ψ), which represent the linear distribution of normal and tangential tractions, and v_f of the cell, resulting in $D = 17$. The ψ coefficients are obtained from the equilibration process performed after the macro-level TO, ensuring that boundary conditions are properly imposed, as illustrated in Figs. 1 and 2.

Each optimised cell's geometry is defined in the space $I \in \mathbb{R}^{N_e \times N_e}$ determined by the regular Cartesian mesh used in the fine discretisation of the cell's square domain. Particularly, in this work, it corresponds to a $N_e = 32$, chosen for consistency with

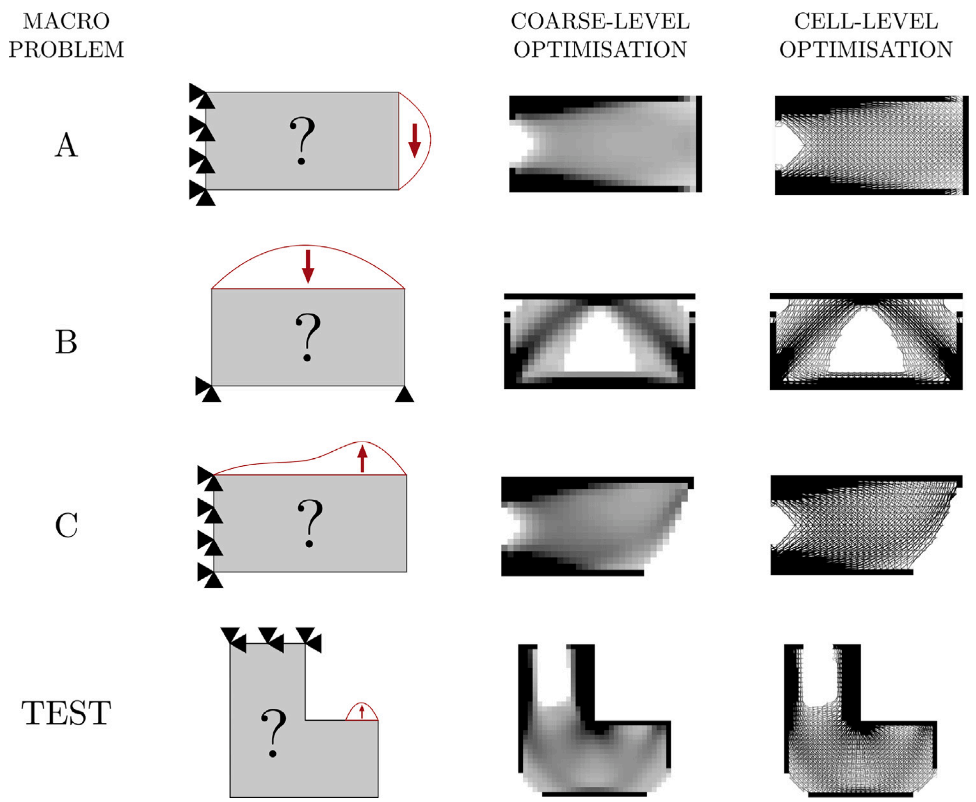


Fig. 3. Description of the global problems used to create the different dataset (Problems A, B and C) and the test problem used to compare between the different numerical analyses.

the Cartesian mesh of previous studies [15]. Both spaces represent the input (μ) and output (I) spaces of the cell's TO process. Consequently, each sample is formed by the information in μ and its mapping in I, obtained by computing multiple iterations of the SIMP method until convergence (see Fig. 2).

However, the dimensionality of μ (D) can significantly increase if tractions distribution is not defined as linear or, especially, if we deal with 3D structural problems. In the latter case, the parameters related to boundary conditions could increase from 16 in the context of two-dimensional problems to 72 in three-dimensional problems, following a similar reasoning.

This increase in dimensionality leads to a *curse of dimensionality*. This phenomenon leads to two main issues. First, the exploration of the input space becomes unaffordable. An effective exploration to create a meaningful dataset requires a prohibitive amount of data in high-dimensional input spaces, which is often impractical or impossible to achieve. Secondly, there is a lack of sensitivity when measuring distances. As dimensionality increases, the distance between any two points becomes almost uniform, and the concept of proximity, distance or nearest neighbour may not even be qualitatively meaningful, diminishing the effectiveness of distance-based algorithms [34].

As a result, it is necessary to propose efficient data creation strategies in the context of the 2-Level TO.

3.2. Dataset creation using efficient strategies

This paper defines a set of global structural problems to obtain multiple optimised cells, which will be the samples of our datasets (see Fig. 3). In particular, three structural problems will be considered to create the available dataset in further sections (Problems A, B and C), while the last problem (2D hook) will be considered as the test problem in all the numerical analyses. Thus, for each structural problem, the 2-Level TO process has been performed, obtaining several optimised cells for each example (see Table 1).

Given the vastness of the input space, fully exploring and sampling it is impractical, particularly in 3D due to the curse of dimensionality. Therefore, the dataset construction in this work reflects a realistic industrial scenario, where macro problems arise dynamically based on specific design needs. As each macro problem is solved, its optimised cells are added to a historical dataset, available by the instance-based model. To replicate this process, we define three macro problems that are progressively incorporated into the dataset while solving a distinct test problem (the Hook problem). This setup demonstrates how increasing dataset similarity accelerates convergence. While alternative dataset construction strategies could be valid, this approach ensures alignment with practical applications, where datasets evolve as new problems are solved.

Table 1
Quantity of samples generated for each macro problem.

Global structural problem	Number of original samples
A	374
B	290
C	320
D (Test)	486

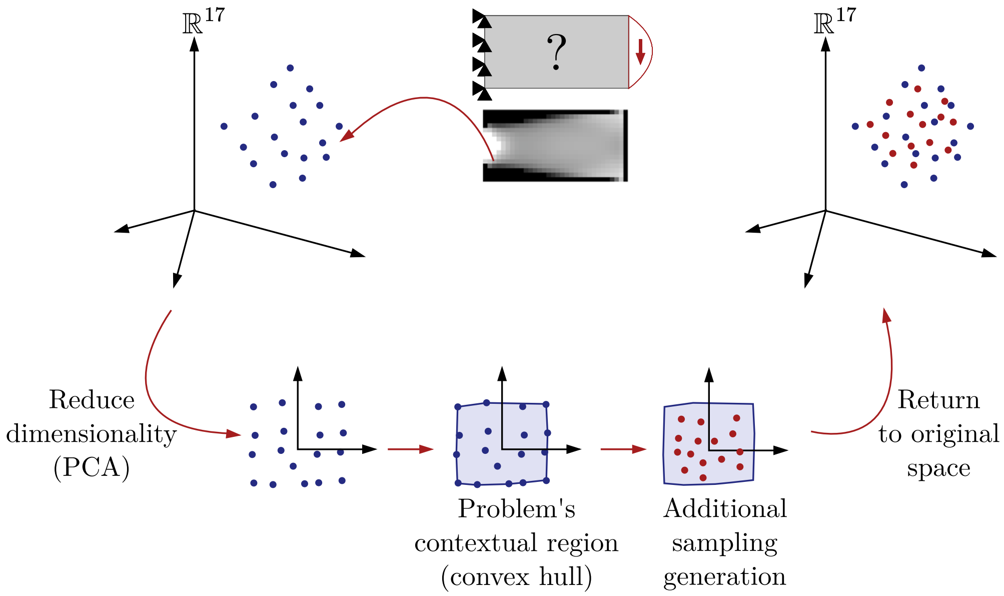


Fig. 4. Workflow scheme to create context-based additional samples: From a global problem's sample distribution in the parameter's space, it is possible to find a manifold by dimensionality reduction. An extra sampling is proposed in a region of interest in that reduced space. Finally, the new samples can be recovered in the original parameter's space and computed TO to include them in the dataset.

Additionally, the dataset creation process involves two strategies. The first strategy focuses on investing computational time on exploring only the regions of interest defined by several global structural problems. This is called the context-based exploration strategy because it is based on information from the coarse discretisation problem (see Section 3.2.1). The second strategy increases the available data related to the stored cells (samples) through data augmentation (see Section 3.2.2). This involves applying simple transformations, such as rotations and flips, to optimised cells, and consequently, to vectors in μ, regardless of the original macro problem.

3.2.1. Context-based exploration strategy for database enrichment

Creating a universal dataset which covers the whole range of possibilities requires a substantial computational effort, and nearly impossible when dimensionality increases. Moreover, in the industrial realm, it may not be necessary to explore the entire parameter space in all contexts, as the applications of the presented methodology could be confined to a set of similar macro-level problems. Hence, we propose an alternative strategy to enrich the available dataset exploring regions of interest in the parameter space according to manifolds defined by individual macro problems.

The proposed strategy must be independently applied to each global problem to generate extra samples for the datasets, as shown in Fig. 4. Each macro problem i is formed by N cells/samples defined by N vectors of 17 parameters in μ. Subsequently, Fig. 5 shows that each group of samples providing independent global problems can be reduced to manifolds with lower dimensionality without losing significant information in this space transformation using PCA. A region of interest can be defined in this reduced space according to the sample distribution. Fig. 6 applies this process to macro problem A as an example. As a first approach, we propose the definition of a convex hull to define a contextualised exploration region. Within this region, we suggest an additional sampling of M samples that belong to the context of macro-level problem i. This additional sampling is performed randomly, though it could also be done with a focus on uniformly distributing the sampling points. Then, those additional samples can be returned to the original space μ with no computational cost. Finally, the optimal geometries of the new M samples are computed using FEM-SIMP and added to the dataset.

Note that this approach does not incorporate an intelligent sampling strategy designed to minimise sampling efforts, such as Bayesian exploration or similar techniques. Instead, the number of new points to be added is predetermined. The context-based sampling strategy ensures that the newly introduced points remain within the manifold of the macro problem, meaning they share similar parameter configurations while still contributing to dataset expansion.

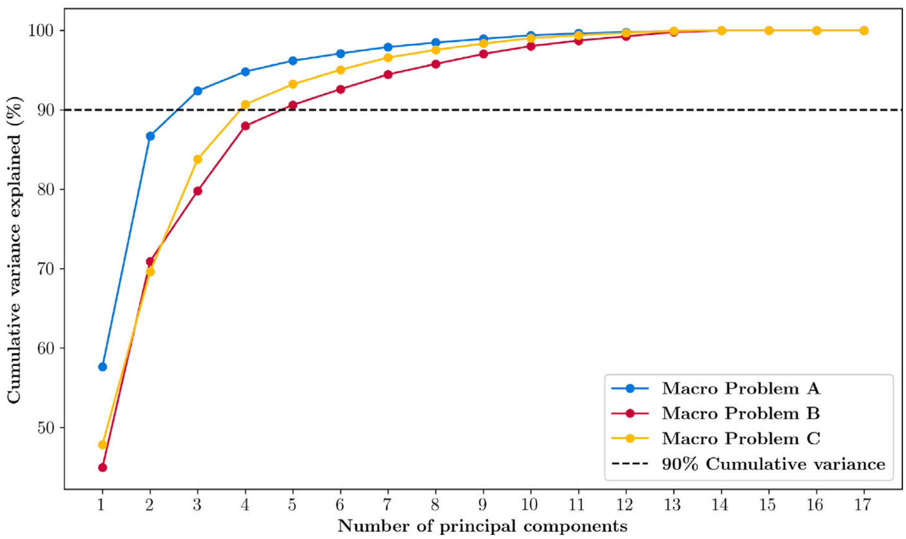
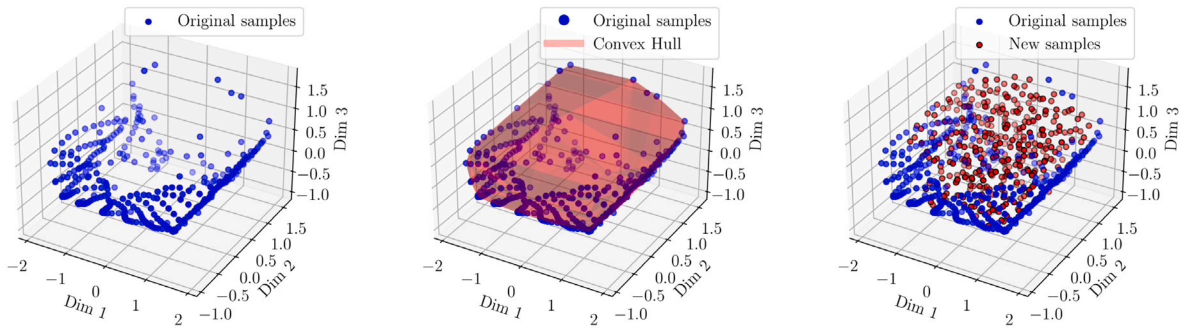


Fig. 5. Cumulative variance explained by principal components from PCA in the different macro structural problems.



(a) Samples distribution in the reduced space obtained from PCA.

(b) Definition of region of interest by a convex hull.

(c) Additional sampling inside the region of interest.

Fig. 6. Application of context-based strategy for data creation to macro problem A. The additional sampling is performed in the subspace obtained from PCA.

3.2.2. Data augmentation strategy for dataset enrichment

In order to increase the available number of cells in our dataset, it is possible to define data augmentation strategies, which are compatible with the previously presented exploration strategy. Data augmentation is a common strategy in deep learning to increase the number of samples by applying some transformations to the original dataset [35]. When dealing with image-based models, a straightforward approach involves applying flips and rotations to the original images. In the particular context of 2-Level TO used in this work, since the geometry space is defined by a density distribution in the square cell discretised by a uniform Cartesian mesh, it is possible to apply rotations and flips to the coordinate axes following the same philosophy.

As a result, for each original sample in the dataset, we combine flipping (up-down and left–right) and rotation (90, 180 and 270 degrees) operations to have the equivalent 8 transformations (see Fig. 7) of each cell. In the same way that transformations are applied to I, the vector in μ must be redefined for each new sample. Thus, seven operations multiply the available data by eight with no additional computational cost.

3.2.3. Dataset creation

Two groups of datasets were created to validate the methods proposed in this paper. The first group was enriched using only the context-based strategy. The second group was enriched by combining the context-based approach with a data augmentation strategy. This allows us to take advantage of the low cost of applying data augmentation to the samples generated from the context-based method.

In both groups of datasets, we adopt an incremental strategy mirroring real-world applications, where multiple macro problems can be gradually added. Thus, three macro problems are introduced to contribute samples to the proposed datasets (see Table 1 and Fig. 3). Dataset Index 1 encompasses samples from macro problem A (see Table 2), dataset Index 2 includes cells from problems A and B, and dataset Index 3 considers problems A, B and C.

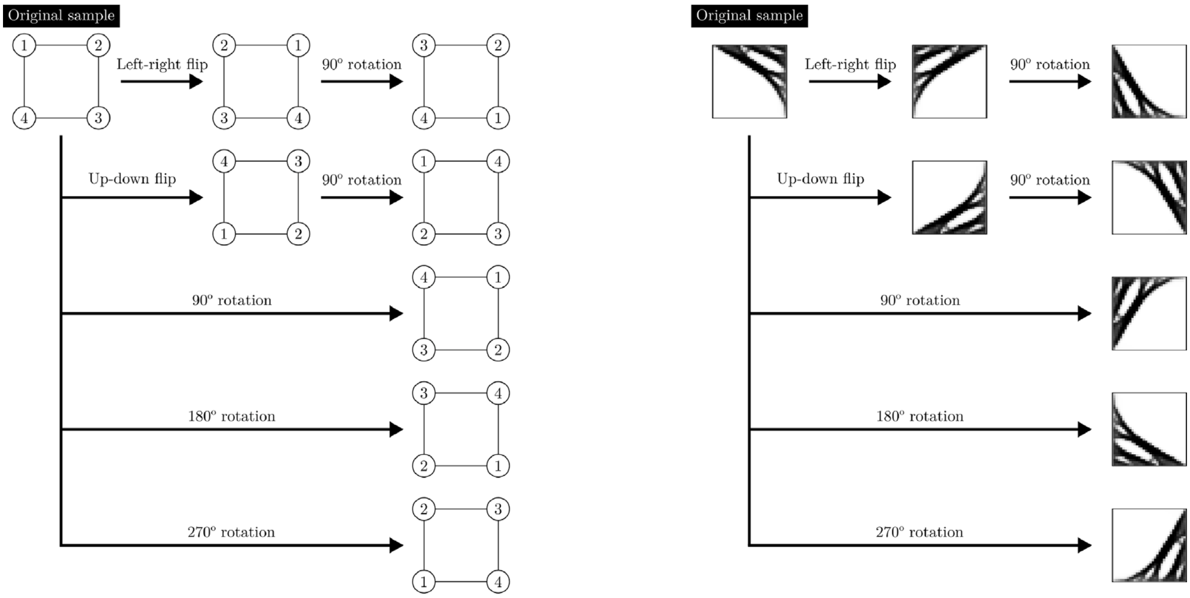


Fig. 7. Transformation operations performed in the data augmentation strategy. Seven operations multiply the available data by eight with no additional computational cost.

Table 2
Number of training samples at each dataset and their origin.

Dataset index	Included global problems	Num. of samples in dataset enriched with context-based strategy	Num. of samples in dataset enriched with context-based and data augmentation strategy
1	A	748	5984
2	A and B	1328	10 624
3	A, B and C	1968	15 744

The number of cells included for dataset Index 1, 2 and 3 for each data creation strategy are shown in the columns of Table 1. First, to dataset index 1, 2 and 3, the context-based enrichment is applied. Thus, the dataset consist of the original n cells from macro problems and the additional m samples derived using the context-based strategy detailed in Section 3.2.1. For the purposes of this study, we determine that m equals n.

Then, we propose a second group of dataset. In this second group, we apply the data augmentation strategy (see Section 3.2.2) to the previous datasets with original and context-based samples. This increases the available information considerably.

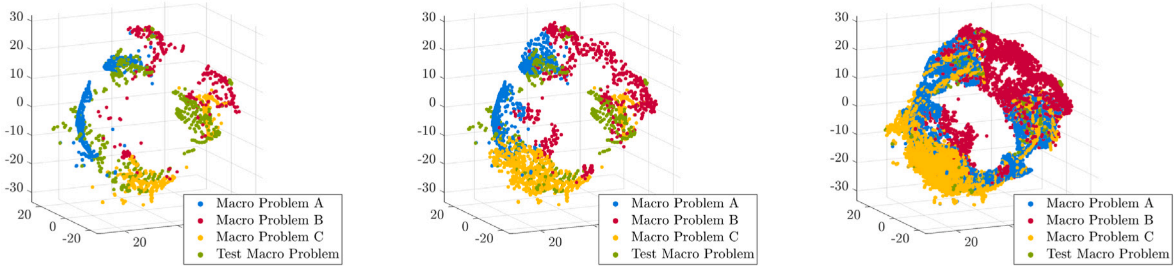
After the several dataset definition, it is interesting to visualise the available data in the two groups of datasets in comparison with the test problem.

Only for visualisation purposes, we use the t-SNE algorithm as a dimensionality reduction technique. The t-SNE [36] algorithm is employed in our study to address the challenges associated with visualising high-dimensional parameter spaces, as it is the case of space μ presented in Section 3.1. In our specific application, t-SNE aids in elucidating the distribution of samples resulting from our intelligent sampling strategy in the space μ, offering insight into the spatial relationships and groupings inherent in the dataset derived from original global structural problems. However, measuring distance in this reduced space is not useful; we just observe the local structure data in clusters since global distances might not be preserved, and it is a non-linear technique.

Consequently, this dimensionality reduction technique is applied to the samples from the largest dataset (Index 3 and both data creation strategies) and the test problem data. Additionally, samples are sequentially visualised in Fig. 8. In Fig. 8(a), only the original samples from global problems are visualised, highlighting the scarcity of data. Then, Fig. 8(b) illustrates the increase of data in the regions of interest following the first data creation strategy. Finally, Fig. 8(c) presents all available information after applying both data creation techniques, showing the enriched dataset.

3.3. Acceleration methodology of the 2-level TO process

As mentioned, the acceleration of the 2-Level TO is focused on the cell-level TO process. The standard approach, presented in [15], of this TO process starts (Initialisation module) with a naive initial guess of the cell's density distribution, which defines a homogeneous density distribution according to the value of cell's v_f. Thus, this approach requires several iterations of the SIMP until



(a) Samples distribution considering only original cells from global problems.

(b) Samples distribution considering original cells from global problems and cells obtained by using context-based strategy.

(c) Samples distribution considering original cells from global problems and cells obtained by using context-based and data augmentation strategy.

Fig. 8. Visualisation of data distribution using t-SNE technique.

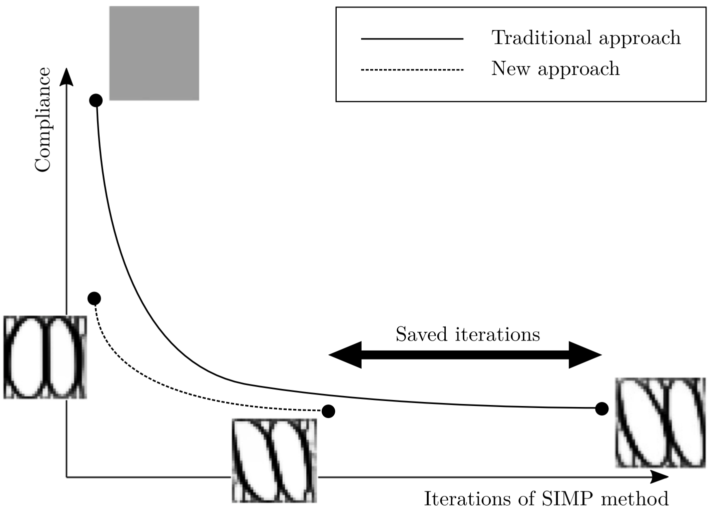


Fig. 9. Comparison of compliance evolution when proposing a homogeneous density distribution and a close initial guess to the local minima.

convergence. However, as in other iterative optimisation processes, the number of iterations until convergence can be significantly reduced if the initial guess is closer to the local minima, the final cell's geometry (see Fig. 9).

To accelerate the process, this work proposes only modifying the TO process initialisation, presented in [15] (see Fig. 1), by selecting the closest sample from a precomputed dataset using an instance-based model (see Fig. 10). Unlike surrogate models such as neural networks, which learn a parametric mapping from inputs to outputs, this approach retrieves solutions directly from the dataset in Section 3.2, requiring no training phase. Given a query cell to optimise, the model selects the nearest available sample, providing a better starting point for optimisation. The main hypothesis is that this method reduces computational cost by requiring fewer iterations to reach the final solution compared to a standard homogeneous distribution. Defining an appropriate similarity metric based on the parameter space, μ, is crucial. Once established, this metric allows the TO process to begin with the most relevant cell density distribution.

3.4. Geometric similarity using metrics in the parameters space

In topology optimisation, the problem can be described by two spaces (see Fig. 11): the input space (X), where parameters defining the problem configuration exist, and the output space (Y), where the solutions reside. These spaces are connected by a function ($\phi : X \rightarrow Y$), where a sample in X represents the design domain, boundary conditions, mechanical properties, and optimisation parameters, and the corresponding solution in Y is the optimal distribution of density, describing the final geometry. The function ϕ represents the optimisation method, such as the SIMP method, and ϕ is not an explicit function in general.

Some authors have studied the relevance of defining metrics based on geometrical features [37] in the context of TO. Those metrics may allow to move through different geometries according to their geometrical features, combining them to get an aggregate feature vector useful for design exploration applications. Their goal is, using the information in the geometry space Y, to find a geometric feature space where the euclidean distance is still meaningful. For that purpose, some dimensionality reduction techniques are applied over Y, which is high dimensional, and finally, acceptable geometric feature spaces have been found.

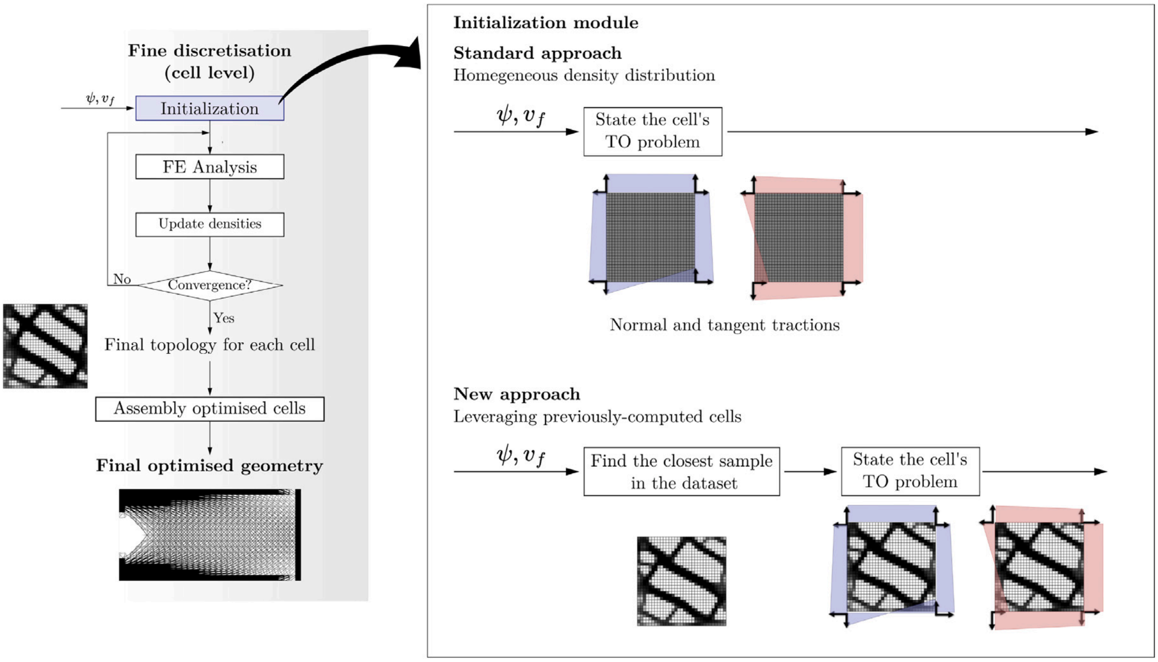


Fig. 10. Comparison of the initialisation module of the standard 2-Level TO method presented in [15] with the new approach introduced in this work. Only the initialisation module of the cell's TO section is modified from the standard method, while the rest of the pipeline remains unchanged. In this new approach, an instance-based model offers a more accurate initial estimate for the query sample, leading to faster convergence with fewer iterations.

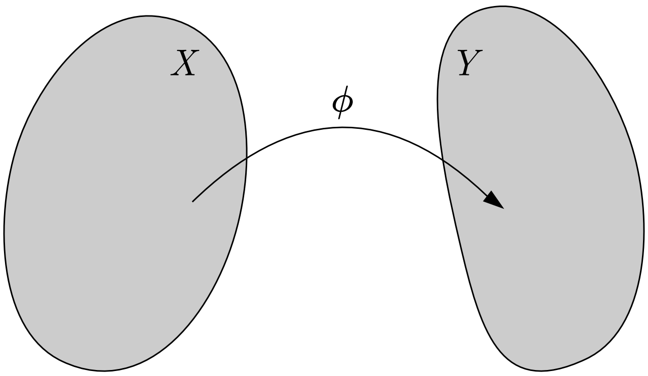


Fig. 11. Input and output spaces mapping.

However, for the purpose of this paper, the main goal is defining a distance to measure similarity between samples (optimised cells) in the input space while considering the mechanical nature of the problem. Thus, in the context of accelerating TO process, the application of those dimensionality reduction techniques might be useful, especially for specific single-scale problems. However, fortunately, we already have a space where the distance is still meaningful in terms of mechanical features: the parameters space μ presented in Section 3.1. These parameters intricately define the TO problem for each cell, since in the context of this work, where each cell domain is the same, the parametric space μ is reduced to the boundary conditions and the prescribed volume fraction.

The challenge is defining a meaningful metric in μ (input space) that is able to provide information on the topology of I (output space). In the surroundings of a given point in μ, μ_0, a topology I_0 exists in I. We assume that in the surroundings of μ_0, little variations around this point will also provoke little variations around I_0 (i.e. locally bounded), which, in our experience, is the most common situation, even if the TO process is generally non-linear.

As a result, we propose two options. First, a vector-based metric where vectors in μ are directly compared using the L1 or L2 norm. Secondly, an energy-based metric, derived from comparing the solutions of the elasticity problem for two samples defined by the parameters in μ with homogeneous density distribution. Both types of distance metrics take into account the mechanical nature of the cell's TO problem. The first approach directly measures differences in the μ parameters that define the TO process, while the energy-based metric is related to the strain energy (compliance) minimised in the TO problem.

3.4.1. Parameter's vector distance

The first approach to compare two samples (μ_i and μ_j) is directly measuring the distance between two parameter's vectors in X. As a result, we define modified L1 and L2 norms as

$$d_{L1\ mod}(\mu_i,\mu_j) = \alpha|v_{f\,i} - v_{f\,j}| + \frac{(1-\alpha)}{K}\left(\sum_k^K \left|\psi_i^k - \psi_j^k\right|\right), \tag{2}$$

$$d_{L2\ mod}(\mu_i,\mu_j) = \alpha(v_{f\,i} - v_{f\,j})^2 + \frac{(1-\alpha)}{K}\left(\sum_k^K \left(\psi_i^k - \psi_j^k\right)^2\right), \tag{3}$$

where v_f is the volume fraction, constrained such that $0 \leq v_f \leq 1$; ψ_i are the parameters related to BC, constrained such that $-1 \leq \psi_i \leq 1$; and α is a weighting factor, constrained such that $0 \leq \alpha \leq 1$, which controls the influence of the two different kinds of parameters. K is the number of BC parameters, which are 16 in the context of two-dimensional problems and 72 for three-dimensional problems.

This increase in dimensionality leads to a curse of dimensionality in the case of 3D structural problems. As a result, although this vector-based distance might be useful in the case of 2D structural problems, it is necessary to propose an alternative metric that is also valid for 3D applications.

3.4.2. Energy-based distance

The SIMP method aims to minimise compliance, which is equivalent to minimising strain energy, defined as

$$\mathrm{U} = \int_V \boldsymbol{\sigma}^T \boldsymbol{\varepsilon}\, dV = \int_V (\varepsilon_x\ \sigma_x + \varepsilon_y\ \sigma_y + \gamma_{xy}\ \tau_{xy})\, dV. \tag{4}$$

In order to compare two samples (μ_i and μ_j), it is essential to consider that components of the applied forces are relevant in terms of TO outcome. For instance, a pure vertical traction load case would result in a distribution of vertical bars, while a pure horizontal traction load case would provide a distribution of horizontal bars, but they would have the same energy (provided the same magnitude in the traction field). As a result, comparing two samples by directly measuring distance in U distribution would not capture those differences since the U distribution is not affected by the load components orientation.

Consequently, this work proposes a new energy-based metric inspired by the sum of components described in Eq. (4). Given the repetitive square cell domain of the cells, we can compare μ_i and μ_j by measuring the product of stress–strain at multiple points in the domain, such as the centre of the element.

Thus, using the parameter's vector of each sample, it is possible to define a naive FE analysis for each sample (uniform density distribution according to v_f) and obtain the solution to the elastic problem, which includes $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$. The forces are defined by ψ, while the cell domain density is a homogeneous distribution of ρ whose value is assigned according to the v_f value of the parameter's vector. Thus, the energy-based distance is defined as

$$\begin{aligned} d_{Energy}\ (\mu_i,\mu_j) = \frac{1}{2}\sum_e^N \Big(&|\varepsilon_{x,e}^i\ \sigma_{x,e}^i - \varepsilon_{x,e}^j \sigma_{x,e}^j| + |\varepsilon_{y,e}^i\ \sigma_{y,e}^i - \varepsilon_{y,e}^j\ \sigma_{y,e}^j| \\ &+ |\gamma_{xy,e}^i\ \tau_{xy,e}^i - \gamma_{xy,e}^j \tau_{xy,e}^j| \Big) V_e, \end{aligned} \tag{5}$$

where N is the number of points of the domain where the distance is evaluated, and V_e is the volume of the element.

Unlike directly measuring the distance of parameter's vectors as in Section 3.4.1, this new distance requires solving a FE analysis to obtain $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ for samples i and j. However, the computational cost is actually negligible due to the use of factorisation strategies since the stiffness matrix is the same in both cases but weighted by the relative density (see Eq. (1b)).

As the influence of volume fraction is already included in the formulation of the elastic problem it is unnecessary to include a weighting factor. Additionally, this metric can be employed without the curse of dimensionality when scaling to 3D structural problems. In that case, the parameter's vector dimensionality scales from 16 dimensions to 72 with this formulation. However, using the energy-based metric ensures that $\boldsymbol{\sigma}$ and $\boldsymbol{\varepsilon}$ scales from 3 dimensions to 6. In this last case, the number of elements in the cell will give a better definition in the compared fields but not in the intrinsic dimensionality of the problem.

Finally, from a mathematical perspective, various fractional L_p-based norms might mitigate the effects of the curse of dimensionality by enhancing distance sensitivity [38], but this study adopts a different approach. We specifically compare the commonly used Euclidean distance (a first-order approximation in L_p spaces) with our proposed energy-based metric. This metric is grounded in the physics of the problem, rendering it more suitable and intuitive within the context of the SIMP algorithm. The implicit relationship between our metric and the strain energy formulation in topology optimisation offers a physically meaningful alternative to purely mathematical distance measures.

4. Results

In this Section, we conduct several numerical analyses to demonstrate how the proposed metrics contribute to accelerating the 2-Level TO process and enhancing data exploration through the use of data creation strategies. Thus, we compare the standard approach presented in [15] with the new approach based on the instance-based model.

First, the vector-based distance metric is applied to generate the initial guess for the TO process. We perform a series of analyses on the two dataset groups introduced in Section 3.2.3 to evaluate the impact of the different data creation strategies. Subsequently, these analyses are repeated using the energy-based metric to compare their respective effects.

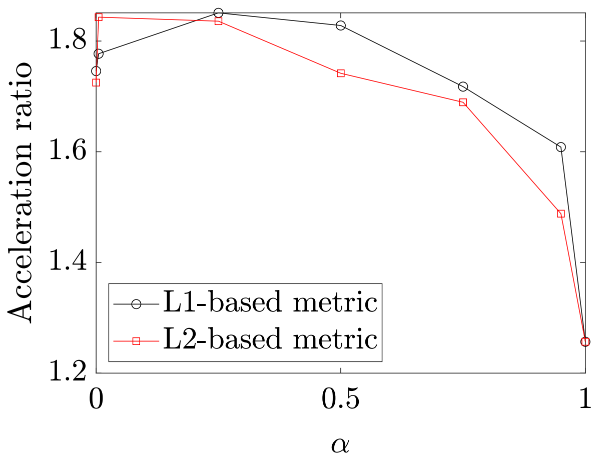


Fig. 12. Influence of metric's weighting factor α in the average acceleration ratio (β) for the test problem.

4.1. Using vector-based metric for speed-up

The definition of the vector-based metric in Section 3.4.1 distinguishes the dimensions according to boundary conditions or volume fraction nature, so it includes the parameter α to control its influence in the metric results. As a result, it is necessary first to conduct a sensitivity analysis of this factor and then choose a proper value to conduct the numerical analyses of the acceleration-distance relationship.

For this purpose, since the main goal of this work refers to the acceleration of the cell's TO process, a new numerical indicator is introduced, denoted as β, to quantify the acceleration ratio. A conventional metric for this purpose is λ, representing the ratio of the number of FEM-SIMP iterations required with a instance-based model (n_{model}) to propose an initial guess, compared to the iterations needed with a homogeneous (naive) initial guess ($n_{standard}$), used in the original methodology presented in [15], as expressed in

$$\lambda = \frac{n_{model}}{n_{standard}}. \tag{6}$$

However, a limitation of λ is its definition within the range $[0, \infty]$, where the acceleration region is [0,1] and the deceleration region is $[1, \infty]$. Thus, it is challenging to compare different phenomena on the same scale. Consequently, we redefine an acceleration ratio β through a second equation, as follows

$$\beta = \begin{cases} \frac{1}{\lambda} & if\ \lambda < 1 \\ -\lambda & if\ \lambda \geq 1. \end{cases} \tag{7}$$

This adjustment shifts the acceleration region to $[1, \infty]$ and the deceleration region to $[-\infty, -1]$, facilitating a more straightforward and comparable interpretation.

4.1.1. Impact of weighting factor α

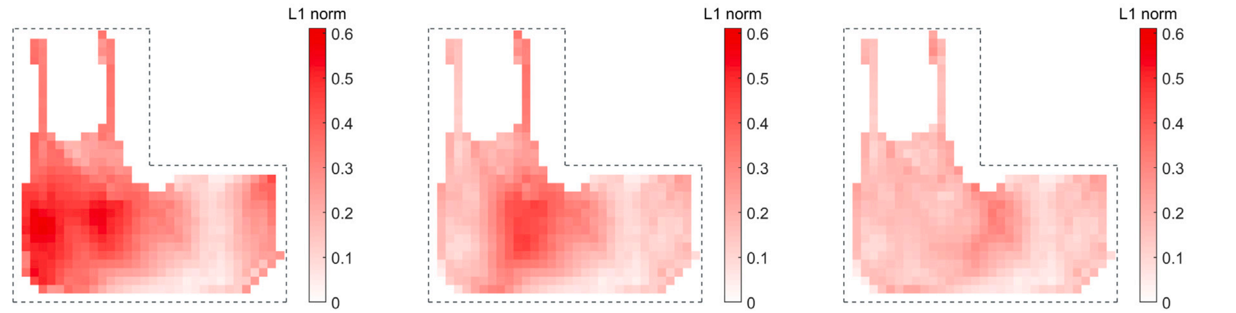
As it is mentioned, α serves as a weighting factor in Eqs. (2) and (3), determining the weight assigned to the volume fraction in comparison to the boundary condition parameters in the distance metric calculation. A modification in this parameter may affect the nearest neighbours for a given point, thereby influencing the initial guess proposed by the instance-based model, and consequently, the acceleration of the process. As a result, choosing a proper value of α directly depends on the acceleration results.

Fig. 12 illustrates the global acceleration results in the test problem presented in Section 3.2, using Dataset Index 3, which is the most populated. Despite in the extreme values of α (0 and 1), the global acceleration remains stable, showing an optimal value at $\alpha = 0.25$. This suggests the importance of considering both types of parameters (v_f and ψ), indicating that, given their similar bounded ranges, the weighting factor is not excessively sensitive.

It is crucial to note that while this study may not fully represent all cases or scenarios, the qualitative behaviour of the weighting factor is expected to be similar in other macro problems and datasets. Specifically, although the optimal value of $\alpha = 0.25$ was identified for Dataset Index 3, it is anticipated that this value will provide a reasonable starting point for other datasets.

Employing a scalar parameter to express the influence of each dimension is a preliminary approach that can be further refined. Advanced techniques, such as neural networks, can be explored to infer the influence of each parameter dynamically and provide optimal weights according to different scenarios.

Despite these considerations, it is important to emphasise that the primary focus of this work is the development of the energy-based metric, which mitigates some limitations associated with traditional L_p-norm metrics. The analysis conducted on the weighting factor α serves to illustrate these challenges and to identify a valid α for the following studies. While exploring the impact of α across the entire input space (using various macro problems or sample sets) poses significant challenges, our findings suggest that the acceleration results remain satisfactory and meaningful, warranting further numerical analyses using the optimal value of $\alpha = 0.25$.

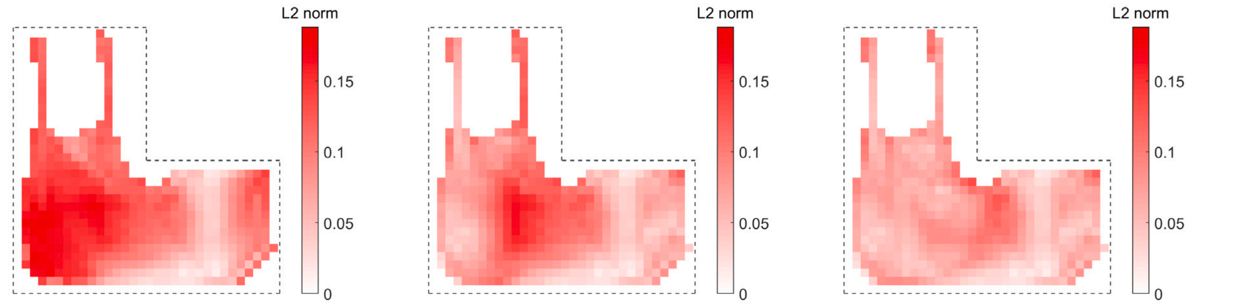


(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 13. Distance to the nearest sample using L_1 norm.

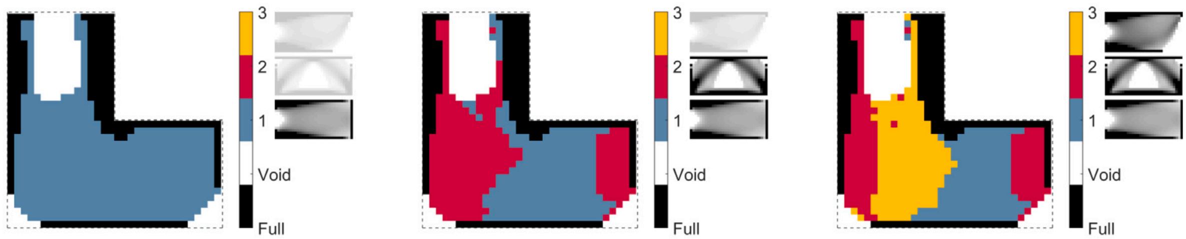


(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 14. Distance to the nearest sample using L_2 norm.



(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 15. Original macro problem of each initial guess of the test problem's cells using L_1 norm.

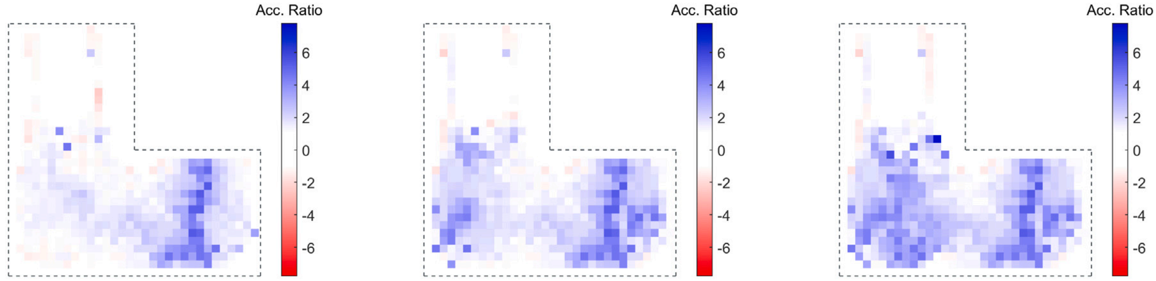
4.1.2. Results from datasets enriched using context-based strategy

Figs. 13 and 14 presents the local distribution of the distance to the nearest neighbour in the dataset for the cells of the test problem using L1-based and L2-based norm (Eqs. (2) and (3)). In the majority of cases, both the L1 and L2 norms exhibit similar behaviour, suggesting the same nearest sample as an starting point for the optimiser. Notably, as the dataset expands with an increase in the number of samples (Index 2 and 3), the mean distance to the nearest neighbour diminishes. This trend becomes more apparent in Fig. 15, where each additional set of samples influences the local distribution of the nearest macro problem. Noteworthy is the observation that samples assigned to problem A in Dataset Index 1 with low distance values persist as the nearest samples, regardless the inclusion of additional samples from problems B and C.

Regarding the local and general perspectives, the relationship between the distance to the nearest neighbour and the acceleration ratio reveals two key aspects. Firstly, aligned with expectations derived from the t-SNE representation of the dataset and the test problem (see Fig. 8), the distance to the nearest neighbour diminishes as the number of available sample in the dataset increases (see Fig. 13 and Table 3).

Table 3
Distance results using the dataset formed by original and contextualised samples.

	L1-based norm		L2-based norm		Energy-based norm	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Dataset index 1	2.1963	1.4775	0.1074	0.0434	0.0045	0.0052
Dataset index 2	1.8210	1.4346	0.0799	0.0343	0.0034	0.0040
Dataset index 3	1.3280	1.4770	0.0638	0.0214	0.0032	0.0038

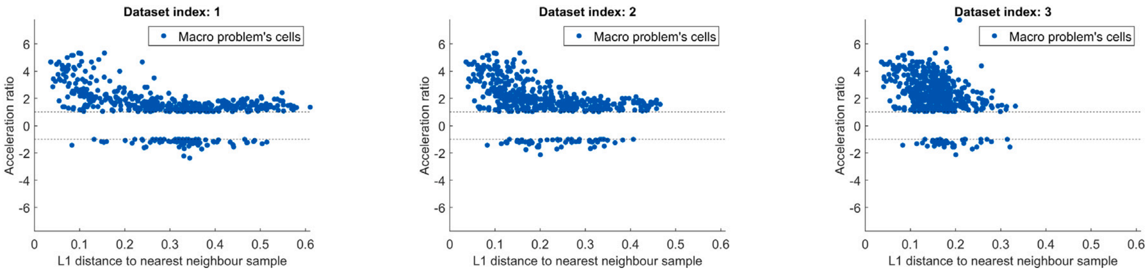


(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 16. Local results of the acceleration ratio using L_1 norm to propose the initial guess.



(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 17. Relationship between acceleration ratio and L_1 distance used to propose the initial guess of the test problem.

Consequently, the second observation is that a higher acceleration ratio is observed when the distance to the nearest neighbour is diminished, as shown in the comparison among Figs. 13, 14 and 16. Thus, reduced distance correlates with increased acceleration. This result is significant because it numerically validates the initial hypothesis that in the vicinity of x_0, the TO process will lead to the vicinity of y_0.

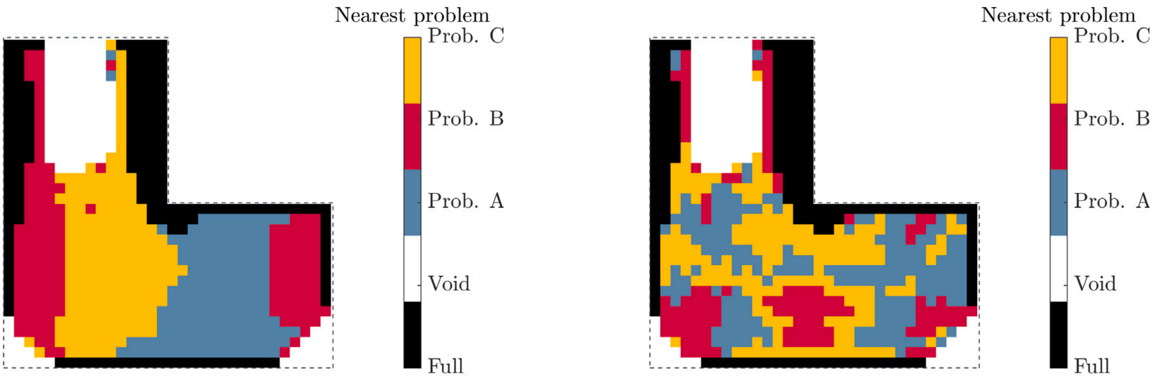
In line with this idea, Fig. 17 illustrates the correlation between both magnitudes. Although some outliers in the deceleration region can be identified, the global trend indicates a reduction in iterations as the distance decreases.

Measuring distances within this space acts as an *a priori* criterion for selecting a nearby initial guess, being related to the acceleration of the cell's TO process leveraging previously computed information. However, using this metric, it is not possible to control the weighting of all the considered dimensions, and the number of dimensions may increase considerably in the context of structural 3D problems.

Despite these limitations, the numerical analyses of incremental datasets underscores the applicability of this approach in real-world scenarios where dataset sizes can be augmented incrementally. Lastly, the context-based strategy facilitates the generation of additional samples without the need to define multiple macro problems, a process that can be both challenging and inefficient.

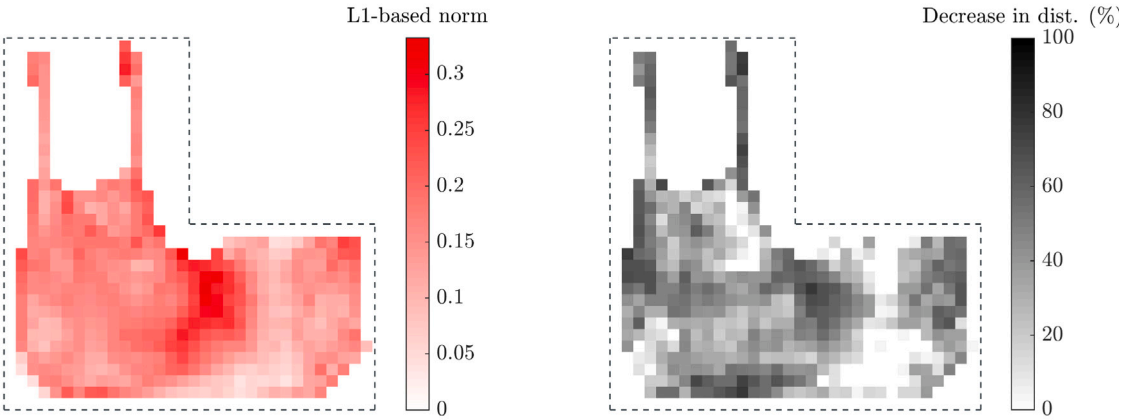
4.1.3. Results from datasets enriched using context-based and data-augmentation strategies

In Figs. 18–20 , we compare the results following the standard strategy without data augmentation versus the data augmentation strategy. This implies a reduction in the distance to the nearest sample in the dataset, as shown in Fig. 19 and Table 4. Finally, as the distance is related to acceleration, this reduction of the mean distance is equivalent to an increase in the acceleration results (see Figs. 20 and 21 and Table 5).



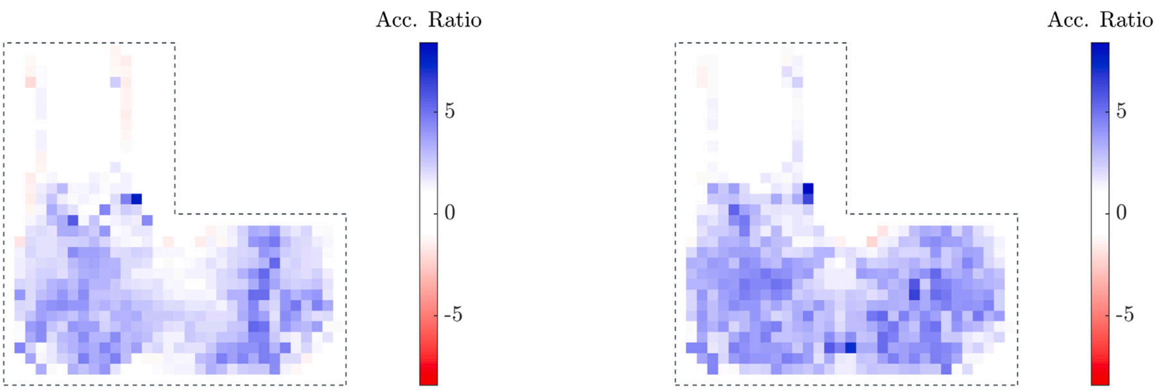
(a) No data augmentation. (b) Data augmentation.

Fig. 18. Comparison of the global problem assigned to the nearest sample when using different strategies.



(a) L_1 distance without using data augmentation strategy. (b) Relative reduction in L_1 distance when using data augmentation strategy.

Fig. 19. L_1-based distance comparison between strategies.



(a) No data augmentation. (b) Data augmentation.

Fig. 20. Acceleration ratio comparison between strategies.

Table 4
Distance results using data augmentation strategy with the dataset formed by original and context-based samples.

	L1-based norm		L2-based norm		Energy-based norm	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
Dataset index 1	0.116	0.044	0.049	0.018	3.41×10^{-3}	4.8×10^{-3}
Dataset index 2	0.103	0.345	0.043	0.143	2.76×10^{-3}	3.50×10^{-3}
Dataset index 3	0.096	0.031	0.040	0.013	2.50×10^{-3}	3.1×10^{-3}

Table 5
Average acceleration results for each numerical analysis using different distance definitions.

	L1-based norm		L2-based norm		Energy-based norm	
	Standard strategy	Data aug.	Standard strategy	Data aug.	Standard strategy	Data aug.
Dataset index 1	1.37	2.43	1.34	2.43	1.39	2.10
Dataset index 2	1.64	2.66	1.61	2.73	1.53	2.38
Dataset index 3	1.85	2.80	1.84	2.86	1.58	2.67

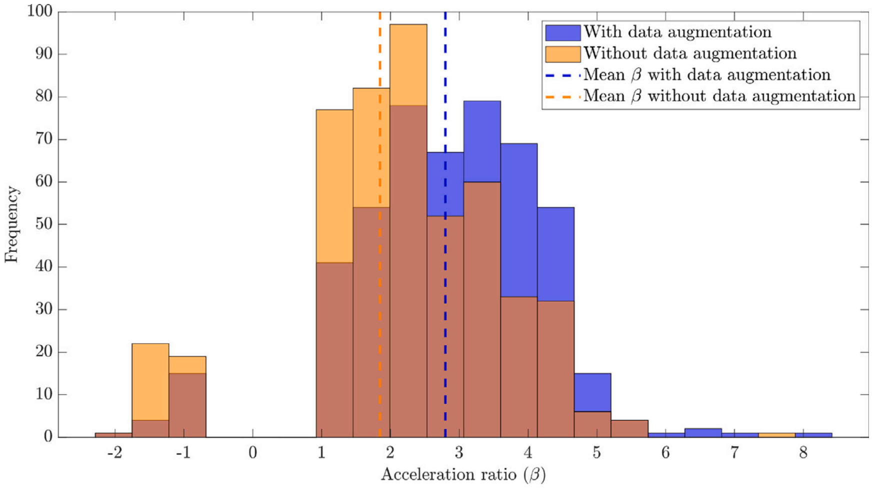


Fig. 21. Histograms of the acceleration ratio for each cell using L_1-based distance, comparing the results with and without the use of a data augmentation strategy.

4.2. Using energy-based metric for speed-up

In this section, we replicate the numerical analyses performed in the previous Sections 4.1.2 and 4.1.3, but using energy-based metric.

4.2.1. Results from datasets enriched using context-based strategy

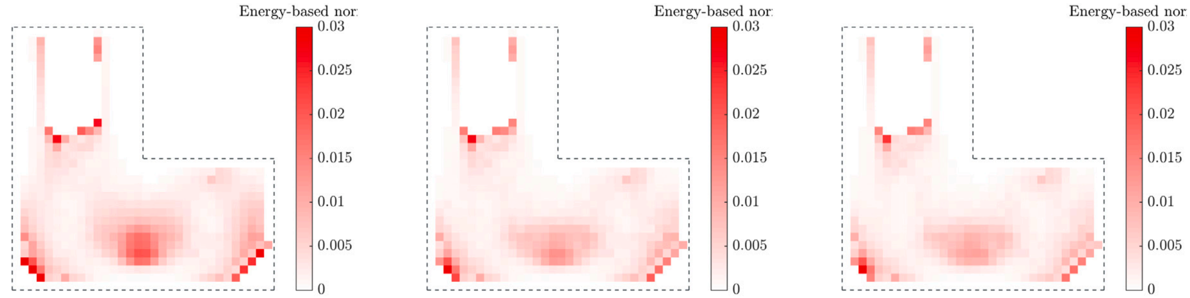
Fig. 22 presents the local distribution of the distance to the nearest neighbour in the dataset for the cells of the test problem using energy-based norm. As in vector-based norms, the distance to the nearest sample is reduced as the dataset information increases. Although the general look of the distance distribution is similar between cases where the L1 and L2-based norms are applied, the distance distribution obtained using energy-based distance is different from the previous approaches and is not similar to them.

Fig. 23 shows the original macro problem of each initial guess of the test problem's cells for this case. Fig. 23(b) (dataset Index 2), and, in particular, Fig. 23(c) (dataset Index 3) show that a more heterogeneous distribution is achieved using the energy based-metric than using L1 and L2-based metrics.

Regarding the acceleration ratio, as shown in Fig. 24, the general trend is similar to the vector-based analyses: the less distance, the higher acceleration. Additionally, the results of mean acceleration ratio are also similar when we compare it with the previous metrics (see Table 5).

4.2.2. Results from datasets enriched using context-based and data-augmentation strategies

The results are presented in Figs. 25–27, and the general trend is similar to the results observed in the numerical analyses using vector-based norm. The distribution of the global problem associated with the nearest sample is heterogeneous, as it is observed when using the standard dataset (see Fig. 25). The distance to the nearest sample is reduced (see Fig. 26 and Table 4), and consequently, the acceleration results significantly improve using the same available data (see Figs. 27 and 28 and Table 5).

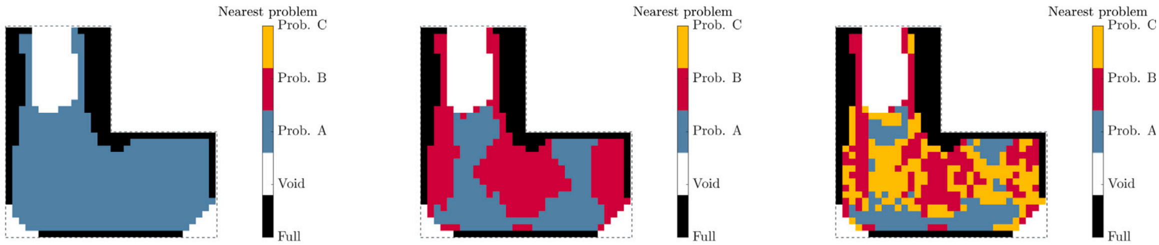


(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 22. Distance to the nearest sample using energy-based norm.

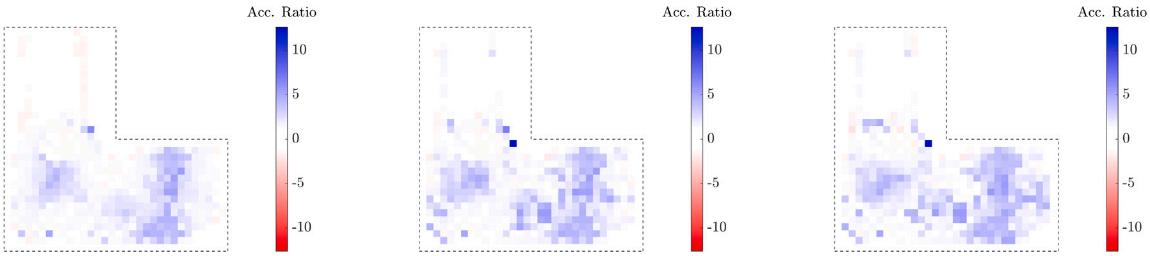


(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 23. Original macro problem of each initial guess of the test problem's cells using energy-based metric.



(a) Test problem solved using Dataset Index 1.

(b) Test problem solved using Dataset Index 2.

(c) Test problem solved using Dataset Index 3.

Fig. 24. Local results of the acceleration ratio using energy-based norm to propose the initial guess.

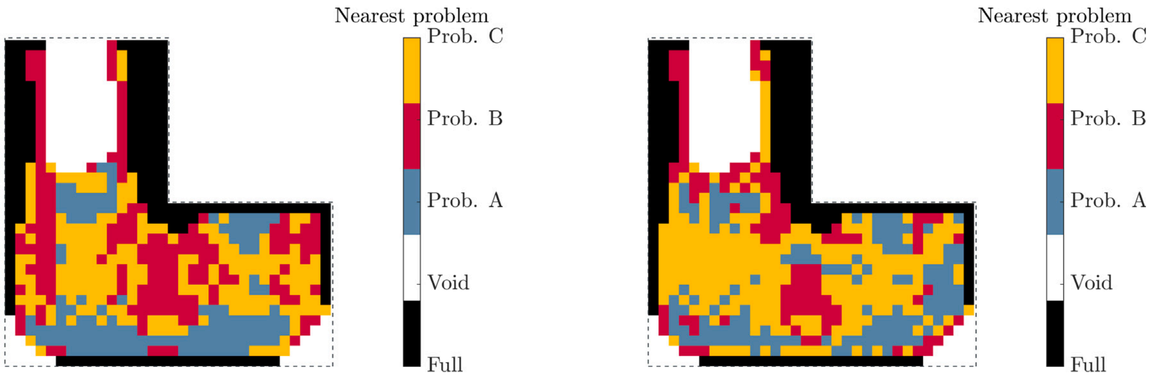
5. Discussion

This section analyses key aspects of the results presented in Section 4. First, Section 5.1 discusses the distance and acceleration results obtained from the multiple analyses presented in Section 4. Section 5.2 presents the characteristics of individual cell optimisation cases, providing insights into the impact of different approaches. Finally, Section 5.3 discusses the final outcome of the macro problem after assembling the optimised cells.

5.1. Relationship between distance and speed-up

The relationship between the proposed distance metric and the observed speed-up is a key aspect of this work. While the distance metric quantifies proximity in the input space, its direct influence on acceleration is not always straightforward, as additional factors contribute to the convergence behaviour of the SIMP method.

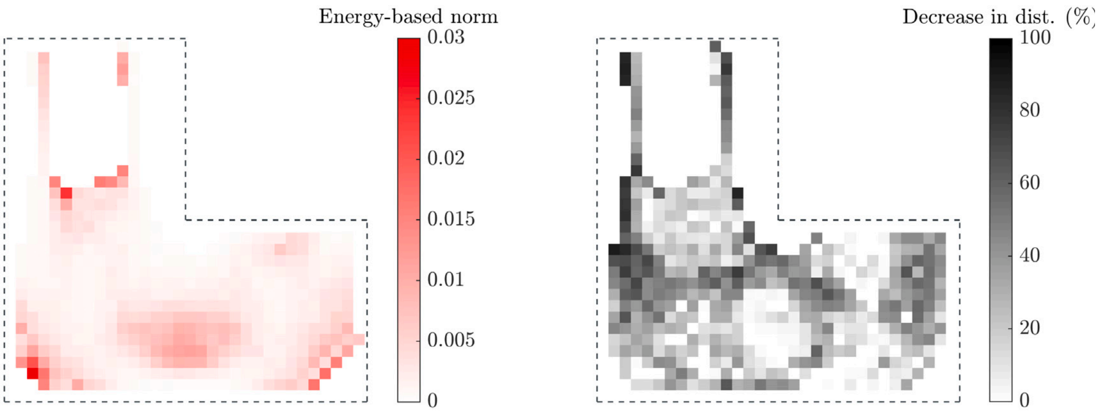
As shown in Table 4, data augmentation introduces only minor variations in the computed distances, suggesting a limited effect on this specific metric. However, the results in Table 5 demonstrate a more pronounced impact on acceleration. This indicates that



(a) No data augentation.

(b) Data augmentation.

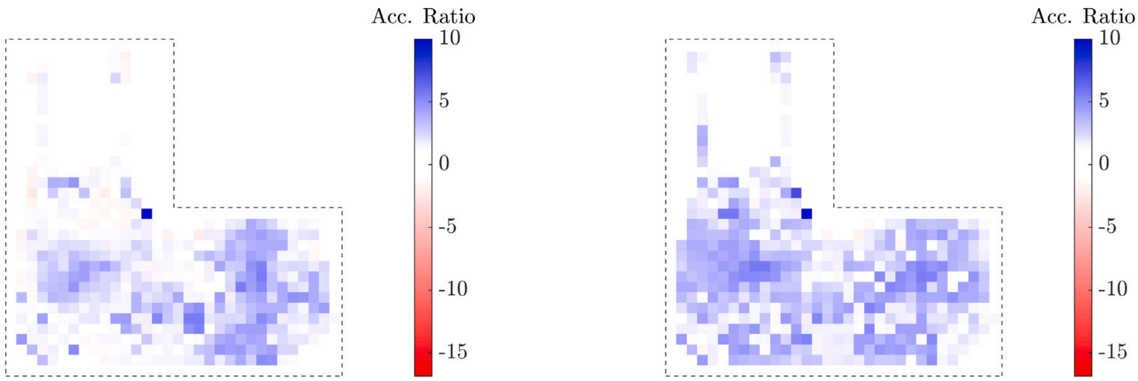
Fig. 25. Comparison of the global problem assigned to the nearest sample when using different strategies.



(a) Energy-based distance without using data augmentation strategy.

(b) Relative reduction in energy-based distance when using data augmentation strategy.

Fig. 26. Energy-based distance comparison between strategies.



(a) No data augmentation.

(b) Data augmentation.

Fig. 27. Acceleration ratio comparison between strategies.

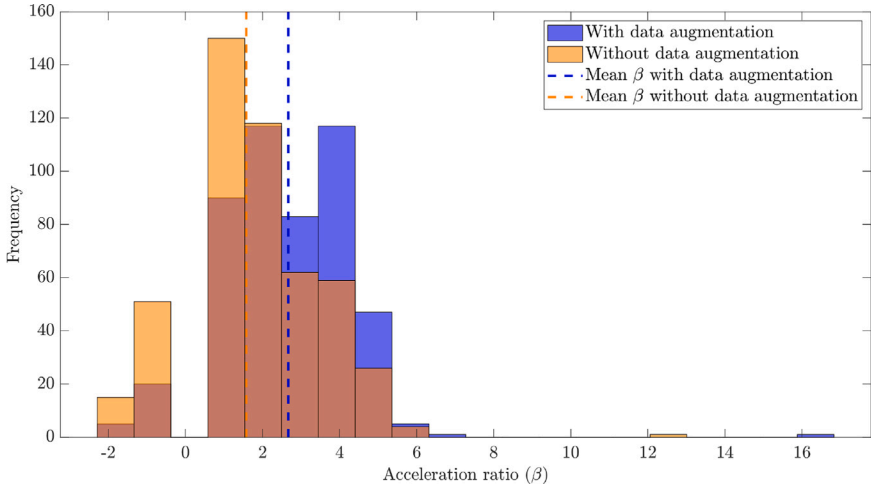


Fig. 28. Histograms of the acceleration ratio for each cell using energy-based distance, comparing the results with and without the use of a data augmentation strategy.

while distance is an important factor, the speed-up achieved also depends on other underlying parameters not explicitly captured by the distance metric.

The acceleration observed in the optimisation process is inherently linked to the number of iterations required for convergence. Since the SIMP algorithm updates the density distribution iteratively, the effectiveness of an initial guess in reducing these modifications plays a crucial role. Despite the complexity of fully characterising all variables involved, these experiments consistently show a trend: smaller distances imply higher acceleration, meaning fewer iterations are required for convergence.

This reinforces the validity of the proposed metric while acknowledging that additional factors influence the optimisation speed. Understanding these interactions remains relevant for further investigation.

5.2. General characteristics of the speed-up results

Figs. 29 and 30 compare the compliance evolution and density distribution for two optimised cells from the Hook Test Problem. For each Figure, the same cell is optimised using the standard approach presented in [15], and the new methodology of this paper. Here, the instance-based model proposes an improved initial guess using Dataset Index 3 and the energy-based metric. These two examples illustrate typical optimisation outcomes observed throughout Section 4.

First, in both cases, starting the optimisation process with an initial guess closer to the final solution reduces the number of iterations required for convergence (see Figs. 29(a) and 30(a)). Notably, in the standard approach, a peak appears in the compliance evolution curves (Figs. 29(a) and 30(a)). This phenomenon may appear when using SIMP method, where the objective function does not always decrease monotonically.

Additionally, the reduced number of iterations in the new approach is visually evident in Figs. 29(b) and 30(b). The initial density distribution in the new approach is already more similar to the final optimised state, requiring fewer updates during the optimisation process.

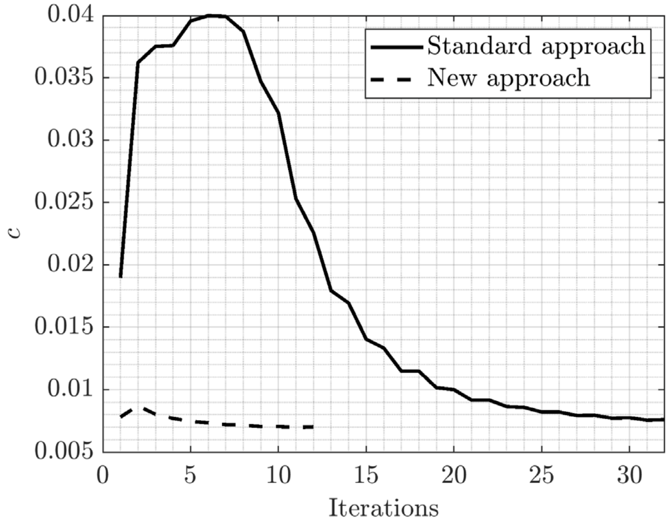
Finally, in terms of the final optimised structure, the achieved compliance values are comparable between both approaches. However, the resulting density distributions exhibit slight differences. This is expected in TO, as multiple local optima exist, and the specific solution obtained depends on various factors, including the optimisation setup. In this case, the variation is directly linked to the use of different initial guesses.

5.3. Comparison between final optimised geometries

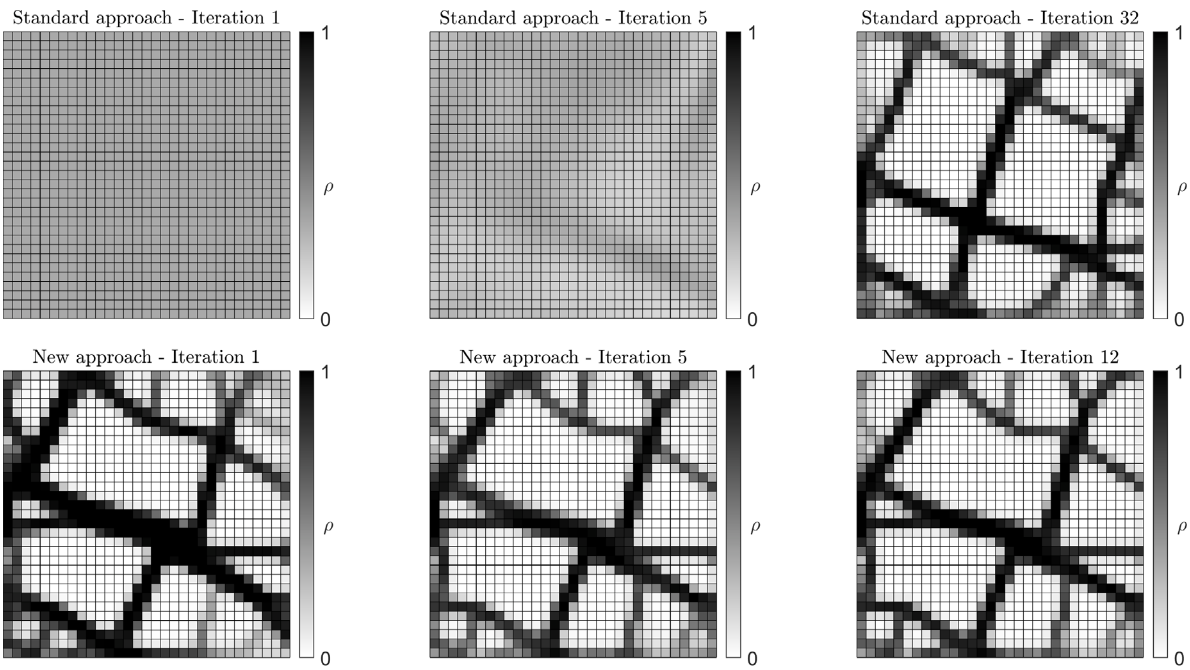
Fig. 31 presents the final geometry of the Hook problem after assembling the individually optimised cells using both the standard approach and the new method proposed in this work. Additionally, a zoomed-in view of a selected region is included to enable a more detailed comparison of adjacent cells.

Fig. 31 shows that, qualitatively and on a global scale, the results are similar. However, the change in the initial guess leads to variations in the optimal distribution of individual cells, particularly in the central region of the square domain, as seen in Section 5.2.

As discussed in detail in [15], the enforcement of equilibrated tractions between adjacent cells ensures smooth intercellular structural continuity, which is also influenced by the mesh resolution. The methodology presented in this paper does not alter this approach, and the SIMP method remains the solver for the individual cell optimisation. Consequently, the results maintain structural continuity and are consistent with the original framework.



(a) Comparison of compliance evolution.



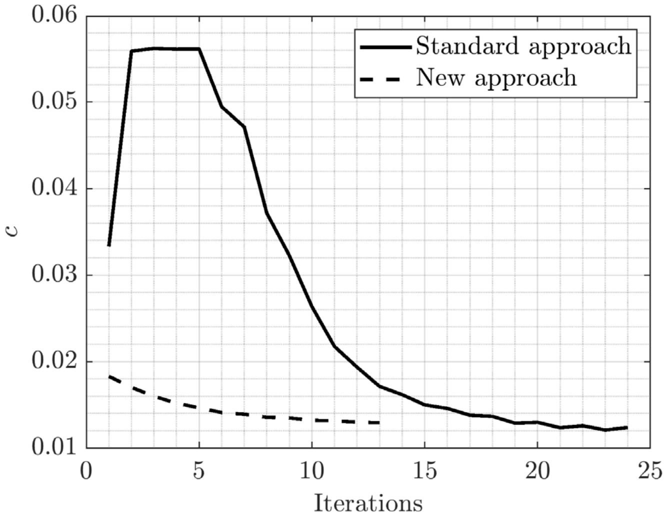
(b) Comparison of the evolution of density distribution.

Fig. 29. Individual analysis for TO process of cell A from Hook Test Problem.

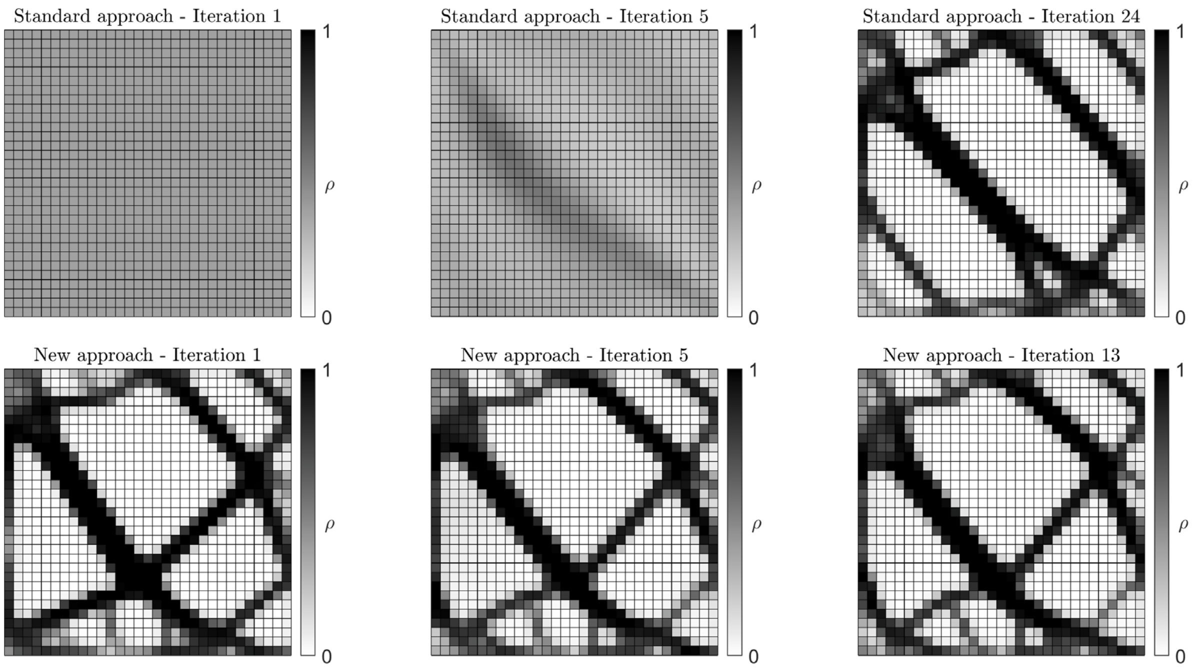
6. Conclusions

This paper presents a novel methodology for accelerating the generation of locally optimised trabecular structures by leveraging previously computed cells. Two data creation strategies are introduced to efficiently build datasets for validating the proposed approach.

The validation uses two metrics, vector-based and energy-based, to enhance the standard cell optimisation process. These metrics provide the SIMP optimiser with a more informed initial guess, positioning it closer to the final solution than the standard uniform density distribution. Both metrics show comparable performance in accelerating the 2-Level TO process across various numerical analyses.



(a) Comparison of compliance evolution.



(b) Comparison of the evolution of density distribution.

Fig. 30. Individual analysis for TO process of cell B from Hook Test Problem.

The data creation strategies, context-based and data augmentation, effectively enrich the dataset for macrostructural problems. As the dataset expands, more relevant samples become available, further speeding up the optimisation process, as demonstrated by t-SNE representations. While the context-based strategy requires solving a cell TO problem for each new sample, the data augmentation strategy is immediate. Combining these strategies improves both the effectiveness and efficiency of data exploration.

The acceleration correlates inversely with the proposed distance metrics, though defining an explicit relationship between them remains challenging. Both metrics fulfil the primary objective of this study, with the choice between them depending on their specific characteristics. As the vector-based metric requires a weighting factor and presents limitations in 3D structural problems, the energy-based metric is the best option.

Additionally, the speed-up strategy performs well even with a limited dataset, highlighting its effectiveness in data-scarce scenarios.

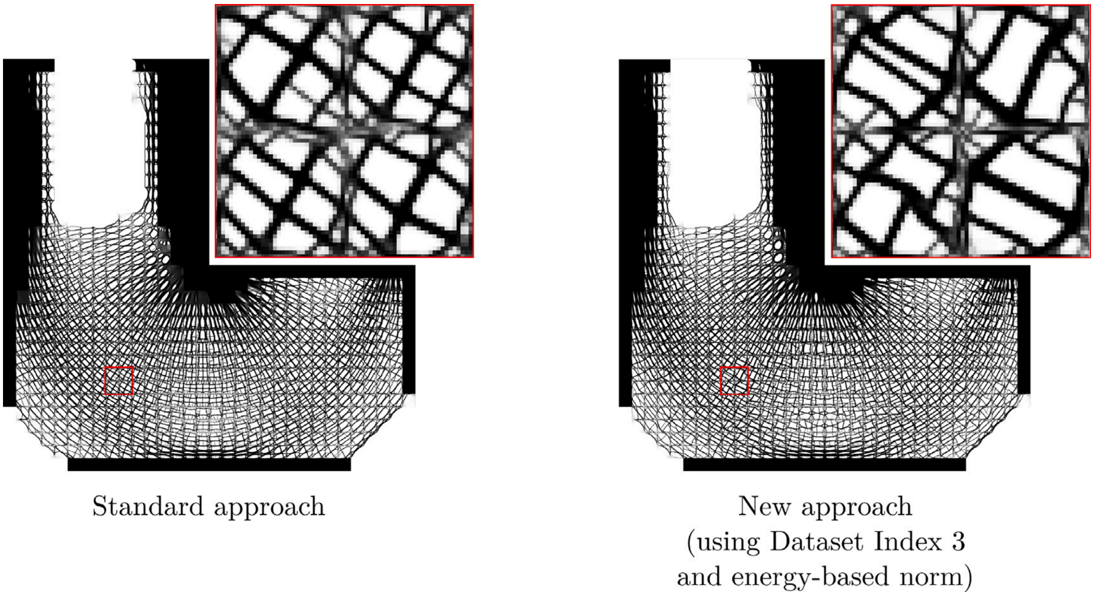


Fig. 31. Comparison of the final optimised geometry using the original strategy presented in [15] and the new approach of an instance-based model using an energy-based metric.

In conclusion, this study demonstrates the potential of the proposed metrics and data creation strategies to improve the efficiency and adaptability of the 2-Level TO process. This advancement is crucial for enhancing the design of high-resolution trabecular structures, particularly for demanding industrial applications.

CRediT authorship contribution statement

A. Martínez-Martínez: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **D. Muñoz:** Validation, Supervision, Methodology, Conceptualization. **J.M. Navarro-Jiménez:** Supervision. **O. Allix:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **F. Chinesta:** Supervision, Methodology, Formal analysis, Conceptualization. **J.J. Ródenas:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **E. Nadal:** Writing – review & editing, Validation, Software, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support of:

- Grant FPU 19/02103 funded by “Ministerio de Educación y Formación Profesional”.
- Grant MCIN/AEI/10.13039/501100011033 funded by “Ministerio de Ciencia e Innovación”.
- Grant PID2022-141512NB-I00 funded by “Fondo Europeo de Desarrollo Regional -FEDER-”.
- Grants Prometeo/2021/046 and CIAICO/2021/226 funded by “Generalitat Valenciana”.

O. Allix would like to thank the French National University Council, ENS Paris-Saclay and National Center for Scientific Research (CNRS) for supporting his sabbatical at UPV, which made it possible to closely interact with the colleagues from I2MB/UPV.

Data availability

Data will be made available on request.

References

[1] O. Ibhadode, Z. Zhang, J. Sixt, K.M. Nsiempba, J. Orakwe, A. Martinez-Marchese, O. Ero, S.I. Shahabad, A. Bonakdar, E. Toyserkani, Topology optimization for metal additive manufacturing: current trends, challenges, and future outlook, Virtual Phys. Prototyp. 18 (1) (2023) e2181192.

[2] A.A. Al-Tamimi, C. Peach, P.R. Fernandes, A. Cseke, P.J. Bartolo, Topology optimization to reduce the stress shielding effect for orthopedic applications, Procedia Cirp 65 (2017) 202–206.

[3] J. Wu, O. Sigmund, J.P. Groen, Topology optimization of multi-scale structures: a review, Struct. Multidiscip. Optim. 63 (2021) 1455–1480.

[4] J. Wu, N. Aage, R. Westermann, O. Sigmund, Infill optimization for additive manufacturing—approaching bone-like porous structures, IEEE Trans. Vis. Comput. Graphics 24 (2) (2017) 1127–1140.

[5] Z. Zhao, X.S. Zhang, Design of graded porous bone-like structures via a multi-material topology optimization approach, Struct. Multidiscip. Optim. 64 (2021) 677–698.

[6] J. Park, T. Zobaer, A. Sutradhar, A two-scale multi-resolution topologically optimized multi-material design of 3D printed craniofacial bone implants, Micromach. 12 (2) (2021) 101.

[7] A. Wubneh, E.K. Tsekoura, C. Ayranci, H. Uludağ, Current state of fabrication technologies and materials for bone tissue engineering, Acta Biomater. 80 (2018) 1–30.

[8] B. Dhandayuthapani, Y. Yoshida, T. Maekawa, D.S. Kumar, Polymeric scaffolds in tissue engineering application: a review, Int. J. Polym. Sci. 2011 (1) (2011) 290602.

[9] Z. Wu, L. Xia, S. Wang, T. Shi, Topology optimization of hierarchical lattice structures with substructuring, Comput. Methods Appl. Mech. Engrg. 345 (2019) 602–617.

[10] G. Allaire, P. Geoffroy-Donders, O. Pantz, Topology optimization of modulated and oriented periodic microstructures by the homogenization method, Comput. Math. Appl. 78 (7) (2019) 2197–2229, Simulation for Additive Manufacturing.

[11] M.P. Bendsoe, O. Sigmund, Topology Optimization: Theory, Methods, and Applications, Springer Science & Business Media, 2003.

[12] M.P. Bendsøe, O. Sigmund, Material interpolation schemes in topology optimization, Arch. Appl. Mech. 69 (1999) 635–654.

[13] N. Aage, E. Andreassen, B.S. Lazarov, O. Sigmund, Giga-voxel computational morphogenesis for structural design, Nature 550 (7674) (2017) 84–86.

[14] A. Ferrer, J. Oliver, J.C. Cante, O. Lloberas-Valls, Vademecum-based approach to multi-scale topological material design, Adv. Model. Simul. Eng. Sci. 3 (2016) 1–22.

[15] R. Merli, A. Martínez-Martínez, J.J. Ródenas, M. Bosch-Galera, E. Nadal, Two-level high-resolution structural topology optimization with equilibrated cells, Computer- Aided Des. 179 (2025) 103811.

[16] Y. Zhang, B. Peng, X. Zhou, C. Xiang, D. Wang, A deep convolutional neural network for topology optimization with strong generalization ability, 2019, arXiv preprint arXiv:1901.07761.

[17] C. Qiu, S. Du, J. Yang, A deep learning approach for efficient topology optimization based on the element removal strategy, Mater. Des. 212 (2021).

[18] G.C. Ates, R.M. Gorguluarslan, Two-stage convolutional encoder-decoder network to improve the performance and reliability of deep learning models for topology optimization, Struct. Multidiscip. Optim. 63 (2021) 1927–1950.

[19] S. Zheng, Z. He, H. Liu, Generating three-dimensional structural topologies via a U-net convolutional neural network, Thin-Walled Struct. 159 (2021).

[20] R. Cang, H. Yao, Y. Ren, One-shot generation of near-optimal topology through theory-driven machine learning, Computer- Aided Des. 109 (2019) 12–21.

[21] D. Patel, D. Bielecki, R. Rai, G. Dargush, Improving connectivity and accelerating multiscale topology optimization using deep neural network techniques, Struct. Multidiscip. Optim. 65 (4) (2022) 126.

[22] D. Bielecki, D. Patel, R. Rai, G.F. Dargush, Multi-stage deep neural network accelerated topology optimization, Struct. Multidiscip. Optim. 64 (6) (2021) 3473–3487.

[23] Z. Zhang, Y. Li, W. Zhou, X. Chen, W. Yao, Y. Zhao, TONR: An exploration for a novel way combining neural network with topology optimization, Comput. Methods Appl. Mech. Engrg. 386 (2021) 114083.

[24] Y. Zhou, H. Zhan, W. Zhang, J. Zhu, J. Bai, Q. Wang, Y. Gu, A new data-driven topology optimization framework for structural optimization, Comput. Struct. 239 (2020).

[25] L. Chen, M.-H.H. Shen, A new topology optimization approach by physics-informed deep learning process, Adv. Sci. Technol. Eng. Syst. J. 6 (2021) 233–240.

[26] H. Jeong, J. Bai, C. Batuwatta-Gamage, C. Rathnayaka, Y. Zhou, Y. Gu, A physics-informed neural network-based topology optimization (PINNTO) framework for structural optimization, Eng. Struct. 278 (2023) 115484.

[27] S. Ye, Y. Lin, L. Xu, J. Wu, Improving initial guess for the iterative solution of linear equation systems in incompressible flow, Math. 8 (1) (2020) 119.

[28] R. Markovinović, J. Jansen, Accelerating iterative solution methods using reduced-order models as solution predictors, Internat. J. Numer. Methods Engrg. 68 (5) (2006) 525–541.

[29] M.P. Bendsøe, Optimal shape design as a material distribution problem, Struct. Optim. 1 (1989) 193–202.

[30] H. Mlejnek, Some aspects of the genesis of structures, Struct. Optim. 5 (1992) 64–69.

[31] D. Muñoz, J. Albelda, J.J. Ródenas, E. Nadal, Improvement in 3D topology optimization with h-adaptive refinement using the cartesian grid finite element method, Internat. J. Numer. Methods Engrg. 123 (13) (2022) 3045–3072.

[32] O. Sigmund, A 99 line topology optimization code written in matlab, Struct. Multidiscip. Optim. 21 (2001) 120–127.

[33] T. Kumar, K. Suresh, A density-and-strain-based K-clustering approach to microstructural topology optimization, Struct. Multidiscip. Optim. 61 (2020) 1399–1415.

[34] C.C. Aggarwal, A. Hinneburg, D.A. Keim, On the surprising behavior of distance metrics in high dimensional space, in: Database Theory—ICDT 2001: 8th International Conference London, UK, January 4–6, 2001 Proceedings 8, Springer, 2001, pp. 420–434.

[35] M. Sonogashira, M. Shonai, M. Iiyama, High-resolution bathymetry by deep-learning-based image superresolution, PLOS ONE 15 (2020) e0235487.

[36] L. Van der Maaten, G. Hinton, Visualizing data using t-sne., J. Mach. Learn. Res. 9 (11) (2008).

[37] N. Dommaraju, M. Bujny, S. Menzel, M. Olhofer, F. Duddeck, Evaluation of geometric similarity metrics for structural clusters generated using topology optimization, Appl. Intell. 53 (2023) 904–929.

[38] C. Aggarwal, A. Hinneburg, D. Keim, On the surprising behavior of distance metric in high-dimensional space, in: Jan Van den Bussche (Ed.), First publ. in: Database theory, ICDT 200, 8th International Conference, London, UK, January 4 - 6, 2001, in: Lecture notes in computer science 1973, Springer, Berlin, 2002, pp. 420–434.